

PD-Prune: Disaggregated Prefill-Decode Pruning for Efficient Agentic LLM Serving

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Abstract

Agentic serving expands one trunk of length L into k residuals and keeps one spine. Disaggregated prefill–decode transfers every computed page, and decode-time pruning can score a child only after that transfer.

PD-Prune admits each residual on the prefill side of the same cut (Figure 1). A full-page hit moves nothing, a partial hit moves the miss, a repeated loop never enters the kernel, and every other residual is prefilled. A low decode score continues past a t_0 -token probe only when the generated prefix meets the threshold. The cut-offs are (0.15, 0.45). The trunk is an alias, so L is absent from the kernel and from the connector.

On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ cuts prefill from 5008 to 3848 tokens at 16/16 any-finished and reduces ESC time from 17.05 s to 14.14 s. On MATH-500 a 128-token probe restores 20/32 at 72.2 s against 104.1 s. The Qwen3-14B spine matches prefix-cache accuracy at $0.72\times$ the median peak KV.

1 Introduction

Large language models improve on hard problems when inference is allowed to search: chain-of-thought writes the intermediate steps, and tree search, self-consistency, and parallel decoding expand several of those steps from one prefix [56, 45, 40, 21]. The accuracy gain is bought with tokens. On GSM8K and MATH, and in agent programs that interleave model calls with tools, most of the expanded branches are not the continuation that is kept [57, 58, 46, 44]. Recent path pruners attack that waste by dropping a branch once a window agrees, a partial reward falls below a quantile, or a learned token marks the prefix as futile [21, 31, 19, 60, 61]. Those tests read text the model has already produced. On a serving stack the prefill, and under disaggregation the KV transfer, have already been paid.

The serving unit is therefore a *branching execution*: one trunk of length L , k residuals ℓ_i , and one committed spine. Three families of systems apply to that execution, and each incurs branch cost at a fixed stage (§2). Automatic prefix caching (APC) [17] and RadixAttention [53] reuse a trunk after a full block has been published, so a same-batch fan-out still materializes k copies of L . Prefill–decode disaggregation [54, 32] isolates TTFT from inter-token latency and then transfers every page the prefiler computed, aborted residuals included. Decode-time pruning scores a child from tokens that are already resident: ESC [21] waits for an agreeing window, Speculative Rejection [31] waits for a partial-reward horizon, and DPTS [19] waits for a mini-step. In each case the residual has already been prefilled. Program-, session-, and workflow-level schedulers [25, 16, 20, 43, 36, 23, 48] raise the scheduling unit above the request and still issue the next call as a unique continuation. Once KV retention has removed trunk recompute [20, 43], the remaining TTFT term is residual-suffix prefill of children the harness can name before the observation returns.

This work. PD-Prune is one admission rule on a shared trunk (Figure 1, §3.1). For residual ρ_i , m_i is the longest page-aligned prefix already in the radix index. That length fixes both the prefill W_i and the transfer X_i : a full hit moves nothing, a partial hit moves the miss, a repeated loop ($i \in \Lambda$) is withheld, and every other miss is prefilled. A score below the decode threshold extends past t_0 tokens only if the probe meets the threshold. The two cutoffs are not the same number. The prefill threshold $\tau_{\text{pre}}=0.15$ reads the residual and drops a loop (0.02) while still starting illegal text (0.22); the decode threshold 0.45 reads the generated prefix, where text that is still illegal scores 0.16 and a finished answer marker scores 0.92 (§3.2).

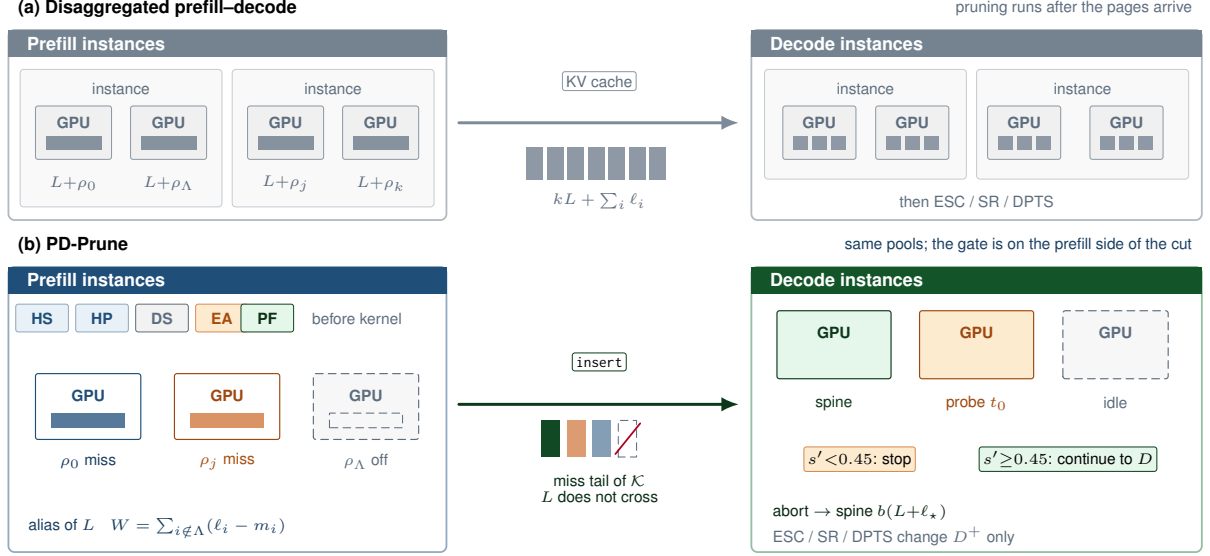


Figure 1: **PD-Prune on a prefill-decode cut (§3.1)**. (a) Stock disaggregation prefills every residual, including a copy of the trunk, and the connector carries $kL + \sum_i \ell_i$. Decode-time pruning runs only after those pages arrive. (b) The two pools stay separate. Admission runs before the prefill kernels: a repeated loop never enters a GPU, a full-page hit moves nothing, and the connector carries the miss tail of $\mathcal{K}$. The trunk is one alias, so L does not cross. Decode extends a t_0 probe only when the prefix meets τ_{dec} ; abort leaves the spine.

Index 0 stays in $\mathcal{K}$, so the connector carries the miss tail of the retained set rather than the k -way trunk [54, 32]. The trunk is aliased at fork (§B.2), which is why L is absent from W_i and from X_i ; abort leaves $M_{\text{spine}} = b(L + \ell_\star)$. Decode slots are $C = \lfloor H/M_{\text{spine}} \rfloor$, and any further prefill uses the budget left after committed decode (§B.6). (8) binds the observed prompt; speculative tokens stay out of the sampler. Where a table separates the alias from the gate, FS is the alias and PD-Prune+ is the same system with the admission rule enabled.

§4 measures that rule. On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ in front of ESC, Speculative Rejection, or DPTS admits 12 of 16 residuals, cuts computed prefill by 23% (5008 to 3848), keeps 16/16 any-finished at $k=8$ and $k=16$, and reduces ESC time from 17.05 s to 14.14 s; $\tau_{\text{pre}}=0.45$ is faster at $k=16$ (10.93 s, still 16/16) and loses two problems at smaller fan-out (Tables 4, 12 and 13). On Qwen3-8B MATH-500, withholding every score below τ drops any-finished accuracy from 20/32 to 13/32 at $k=16$; a 128-token probe restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s (Table 5). Across three turns, 48 of 192 residuals are prefilled and expand time is $7.9\text{--}10.6\times$ below APC (Tables 6 and 15). The Qwen3-14B spine matches APC accuracy at $0.72\times$ median peak KV, and six model families stay in $0.72\text{--}0.75\times$ ($n=40$).

Our contributions are:

- We treat one fan-out as a trunk, k residuals, and a spine, and show that prefix caching, prefill-decode disaggregation, and decode-time pruning each pay for non-winners before a residual can be rejected (§2).
- We introduce PD-Prune, which aliases the trunk, assigns each residual one action that sets both the prefill and the connector payload, and uses separate prefill and decode thresholds so that only a repeated loop is dropped before the kernel (§3).
- On Qwen3-8B and Qwen3-14B we show that the same gate, placed in front of ESC, Speculative Rejection, and DPTS, cuts computed prefill by 23% at matched any-finished accuracy, and that a short probe recovers the accuracy a blanket prefill drop loses (§4).

2 Formulation

A branching step is the object PD-Prune admits. §A is the reason prefix caching, disaggregation, and decode-time pruning each pay for a non-winner before that admission can run; Tables 1 and 7 record the stage and the symbols.

Definition 1 (Branching step). A parent u holds a committed trunk x_u with $|x_u|=L$. A fan-out is a set of residuals $\{\rho_i\}_{i=0}^{k-1}$, $|\rho_i|=\ell_i$. The bound winner $\star$ leaves the spine $x_\star = x_u \parallel \rho_\star$, $|x_\star|=L+\ell_\star$.

Let b be KV bytes per token. The three occupancies of one step are

$$\begin{aligned} M_{\text{clone}} &= b(kL + \sum_i \ell_i), \\ M_{\text{CoW}} &= b(L + \sum_i \ell_i), \\ M_{\text{spine}} &= b(L + \ell_\star). \end{aligned} \quad (1)$$

M_{clone} is the first-expand footprint of recompute, APC, and baseline disaggregation. M_{CoW} is one shared trunk with every residual live. M_{spine} is the footprint after abort. On a cold radix the same span is what disaggregation transfers,

$$X_{\text{PD}} = W_{\text{APC}} = kL + \sum_i \ell_i, \quad (2)$$

because a block hash exists only after the tokens do, and the connector moves every page the prefiller computed [17, 54, 32]. W is the token count issued to a prefill kernel before $\star$ is known. X is the token count inserted on that connector. The admission rule below replaces both with the miss tail of the residuals it keeps, so L is absent from W and from X .

2.1 Where the Fan-out Is Paid

Four mechanisms already sit on this step, and each one can reject a child only after a fixed stage (Table 1). Recompute clones the trunk. APC and a radix index share a block only after that block has been published, so the first fan-out in a batch still materializes k copies of L ; a later replay of the winner is a hit. Prefill–decode disaggregation transfers the pages the prefiller has already computed, aborted residuals included, which is why $X_{\text{PD}} = W_{\text{APC}}$ on a cold index. ESC, Speculative Rejection, and DPTS score a generated prefix, so the residual prefill, and under disaggregation the transfer, precede the drop. PD-Prune aliases one trunk, withholds a repeated loop before the kernel, and continues every other low score only through the probe in (11). The connector then carries the miss tail of $\mathcal{K}$, not the k -way clone.

3 Method

PD-Prune is one first-match action per residual. The action fixes the prefill, the connector payload, and the decode length. §B is the copy-on-write pool, the radix picture, and the span table that realize the same quantities.

3.1 One Action per Residual

Let H be the published radix and P the page size. Only full pages enter H [17]. The page-aligned

match, the repeated-loop set, and the early-abort bit are

$$\begin{aligned} m_i &= \max\{jP : jP \leq \ell_i, \rho_i[0:jP] \in H\}, \\ \Lambda &= \{i \neq 0 : \text{Loop}(\rho_i)\}, \\ F_i &= [\text{Score}_{\text{pre}}(\rho_i[0:\lceil \alpha \ell_i \rceil]) < \tau_{\text{pre}}]. \end{aligned} \quad (3)$$

The trunk is an alias, $\sigma_i \leftarrow (\sigma_u, |x_u|)$, so $L \notin W_i$. Index 0 is forced into the retained set $\mathcal{K}$, and therefore $a_0 \in \{\text{HS}, \text{HP}, \text{PF}\}$.

Definition 2 (Residual action). With $\mathcal{A} = \{\text{HS}, \text{HP}, \text{DS}, \text{EA}, \text{PF}\}$, the first match is

$$a_i = \begin{cases} \text{HS} & \text{if } m_i = \ell_i, \\ \text{HP} & \text{if } 0 < m_i < \ell_i, \\ \text{DS} & \text{if } m_i = 0 \wedge i \in \Lambda, \\ \text{EA} & \text{if } m_i = 0 \wedge i \notin \Lambda \wedge F_i, \\ \text{PF} & \text{if } m_i = 0 \wedge i \notin \Lambda \wedge \neg F_i. \end{cases} \quad (4)$$

HS is a full-page hit, HP a partial hit, DS a repeated loop withheld before the kernel, EA an early abort at fraction α of the residual, and PF a full prefill. The Iverson expansion of the same first match is (18). The lengths are

$$W_i = \begin{cases} 0 & \text{if } a_i \in \{\text{HS}, \text{DS}\}, \\ \ell_i - m_i & \text{if } a_i = \text{HP}, \\ \lceil \alpha \ell_i \rceil & \text{if } a_i = \text{EA}, \\ \ell_i & \text{if } a_i = \text{PF}, \end{cases} \quad (5)$$

$$X_i = W_i [i \in \mathcal{K}] [a_i \in \{\text{HP}, \text{PF}\}]. \quad (6)$$

Summing over the fan-out gives the kernel, the connector, and the decode budget

$$\begin{aligned} W &= \sum_{i=0}^{k-1} W_i, & X &= \sum_{i=0}^{k-1} X_i, \\ D^+ &= \sum_{i=0}^{k-1} d_i. \end{aligned} \quad (7)$$

$X_i=0$ on HS, DS, and EA: a hit is already on the decoder, a withheld loop is never inserted, and an early abort is not transferred. Baseline disaggregation moves (2); the rule moves (6).

On prompt x at u , commit binds the observed continuation and aborts every sibling,

$$v^\star = \arg \max_v \text{LCP}(x, x_v), \quad g_u \leftarrow g_u + 1. \quad (8)$$

Pages of the LCP become the next trunk. A job tagged with a stale generation drops in $O(1)$. Speculative tokens stay outside the sampler, so the emitted tokens match a baseline that prefills committed tokens only.

Table 1: Where each method pays for one fan-out of k residuals on a trunk of length L . “First expand” is the GPU prefill issued before any child has published a block. “Rejection” is the earliest point at which a non-winner can leave. The PD-Prune row is the action of §3.1.

Method	First expand	Rejection	Connector
Recompute	k clones of L	end of the request	none
APC, radix, hash prefill	k clones of L	LRU after a block is published	none; a later replay of the winner is a hit
P/D disaggregation	k clones of L	after insert	$kL + \sum_i \ell_i$
ESC, Speculative Rejection, DPTS	every residual	after a generated prefix	KV of every prefilled child
PD-Prune	one alias of L	loop before the kernel; other low scores after a t_0 probe	miss tail of $\mathcal{K}$

Lemma 1 (First-fan-out dominance). *On a cold index, $W_{\text{APP}} \leq W_{\text{CoW}} \leq W_{\text{APC}} = W_{\text{PD}}$ and $X_{\text{APP}} \leq X_{\text{PD}}$. The inequality on W is strict whenever at least one repeated loop is withheld, and the inequality on X is strict whenever $\mathcal{K} \neq [k]$ or any kept residual is a hash hit.*

Proof. A cold index has $m_i=0$, so (4) never returns HS or HP. Aliasing the trunk removes the k copies of L from W_{APC} , which leaves $W_{\text{CoW}} = L + \sum_i \ell_i$ against $W_{\text{APC}} = kL + \sum_i \ell_i$. (5) then drops every $i \in \Lambda$ and charges $\lceil \alpha \ell_i \rceil$ on EA, so $W_{\text{APP}} \leq \sum_i \ell_i \leq W_{\text{CoW}}$ after the shared trunk is counted once. (6) sums W_i only over $\mathcal{K} \cap \{\text{HP}, \text{PF}\}$, which is a subset of the pages (2) transfers. $\square$

The resulting wall-clock of one fan-out is

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{CoW}} + T_{\text{abort}}, \quad (9)$$

where T_{CoW} is a pointer swap and T_{abort} frees residual pages only. Table 8 compares the six spans.

3.2 Separate Prefill and Decode Thresholds

The prefill score and the decode score read different strings, so the thresholds are a pair $(\tau_{\text{pre}}, \tau_{\text{dec}})$ rather than one cutoff reused at both stages. A blanket prefill drop,

$$d_i^\times = D \cdot [i = 0 \vee s_i \geq \tau_{\text{dec}}], \quad (10)$$

sets $d_i^\times=0$ for every low score and never runs the kernel on that residual. (11) replaces that rule outside Λ .

$$d_i = \begin{cases} 0 & i \in \Lambda, \\ D & i=0 \text{ or } s_i \geq \tau_{\text{dec}}, \\ t_0 & s'_i < \tau_{\text{dec}}, \\ D & s'_i \geq \tau_{\text{dec}}. \end{cases} \quad (11)$$

where $y_i = \text{Dec}(x_u \| \rho_i, t_0)$ is a probe of t_0 tokens and $s'_i = \text{Score}_{\text{dec}}(y_i)$. Loops never enter the kernel. Every other residual is prefilled. A residual

Table 2: Draft scores on the two strings the thresholds read. (12) holds at $\tau_{\text{pre}}=0.15$ because a loop is 0.02 and illegal text is 0.22. (13) holds at $\tau_{\text{dec}}=0.45$ because a still-illegal prefix is 0.16 and an answer marker is 0.92.

	loop	illegal	live text
$\text{Score}_{\text{pre}}, \text{residual}$	0.02	0.22	1.00
$\text{Score}_{\text{dec}}, \text{probe}$	0.04	0.16	0.92

whose prefill score lies below τ_{dec} continues from t_0 to D only when the probe itself clears τ_{dec} . Index 0 is in the second case, so the spine is never probed away.

Proposition 2 (Threshold separation). *Let $s_i = \text{Score}_{\text{pre}}(\rho_i)$ and $s'_i = \text{Score}_{\text{dec}}(y_i)$. The pair $(\tau_{\text{pre}}, \tau_{\text{dec}})$ withholds loops before the kernel and extends a probe only after an answer marker when*

$$\sup_{i \in \Lambda} s_i < \tau_{\text{pre}} \leq \inf_{i \notin \Lambda, i \neq 0} s_i, \quad (12)$$

$$\sup_{y \text{ illegal}} s'(y) < \tau_{\text{dec}} \leq \inf_{y \text{ marked}} s'(y). \quad (13)$$

Table 2 is the measured witness for Theorem 2. The operating point is $(\tau_{\text{pre}}, \tau_{\text{dec}}) = (0.15, 0.45)$. Swapping the two numbers breaks one of the inequalities. Using 0.45 as τ_{pre} puts illegal text (0.22) on the same side of the cutoff as a loop, which is (10) and drops the residual before the probe exists. Using 0.15 as τ_{dec} leaves a still-illegal prefix (0.16) above the line, so the probe runs out to D . Algorithm 1 is the pair: τ_{pre} enters only through Λ and F_i , and τ_{dec} enters only through d_i .

Proposition 3 (Host-invariant prefill). *On a cold index, with $F_i=0$ for every $i \notin \Lambda$,*

$$W(\text{host}) = \sum_{i \notin \Lambda} \ell_i \quad (14)$$

for every decode-time host that reads a generated prefix. The host changes D^+ , not W .

Proof. $m_i=0$ and $i \notin \Lambda$ and $\neg F_i$ force $a_i=\text{PF}$ in (4), so $W_i=\ell_i$. Every $i \in \Lambda$ takes DS and contributes 0. ESC, Speculative Rejection, and DPTS

Algorithm 1 PD-Prune admission. H stores full pages only. Λ is the repeated-loop set. $0 \in \mathcal{K}$ always. (11) sets d_i ; (10) would have set $d_i=0$ for every $s_i < \tau_{\text{dec}}$.

Input: $\{\rho_i\}_{i=0}^{k-1}$, radix H , page P , $(\tau_{\text{pre}}, \tau_{\text{dec}})$, probe t_0 , budget D , fraction α
Output: a_i, W_i, X_i, d_i , and $\mathcal{K}$

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1:  $\Lambda \leftarrow \{i \neq 0 : \text{Loop}(\rho_i)\}$ 
2: for  $i = 0$  to  $k - 1$  do
3:    $m_i \leftarrow \max\{jP : jP \leq \ell_i, \rho_i[0:jP) \in H\} \{(3)\}$ 
4:    $s_i \leftarrow \text{Score}_{\text{pre}}(\rho_i)$ 
5:    $F_i \leftarrow [\text{Score}_{\text{pre}}(\rho_i[0:\lceil \alpha \ell_i \rceil]) < \tau_{\text{pre}}]$ 
6:   if  $m_i = \ell_i$  then
7:      $a_i \leftarrow \text{HS}$ 
8:   else if  $0 < m_i < \ell_i$  then
9:      $a_i \leftarrow \text{HP}$ 
10:  else if  $i \in \Lambda$  then
11:     $a_i \leftarrow \text{DS}$ 
12:  else if  $F_i$  then
13:     $a_i \leftarrow \text{EA}$ 
14:  else
15:     $a_i \leftarrow \text{PF}$ 
16:  end if
17:   $W_i$  by (5)
18:  if  $i \in \Lambda$  then
19:     $d_i \leftarrow 0$ 
20:  else if  $i = 0 \vee s_i \geq \tau_{\text{dec}}$  then
21:     $d_i \leftarrow D$ 
22:  else
23:     $y_i \leftarrow \text{Dec}(x_u || \rho_i, t_0); s'_i \leftarrow \text{Score}_{\text{dec}}(y_i)$ 
24:     $d_i \leftarrow D[s'_i \geq \tau_{\text{dec}}] + t_0[s'_i < \tau_{\text{dec}}]$ 
25:  end if
26:   $X_i \leftarrow W_i[i=0 \vee a_i \neq \text{DS}] [a_i \in \{\text{HP}, \text{PF}\}]$ 
27: end for
28:  $\mathcal{K} \leftarrow \{0\} \cup \{i : a_i \neq \text{DS}\}$ 
29:  $W \leftarrow \sum_i W_i; X \leftarrow \sum_i X_i; D^+ \leftarrow \sum_i d_i$ 

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are applied to tokens that (11) has already admitted, after W_i has been issued. $\square$

Theorem 3 is why those three hosts, run alone, report the same prefill, and why one τ_{pre} moves that column for all three (§4).

On a cold index every $m_i=0$, and the action step collapses to the probe listing Algorithm 2.

Reading one cold fan-out. With $m_i=0$, (4) returns only DS, EA, or PF. The three prefill scores in Table 2 then fix the action before any token is generated. A loop scores 0.02. That lies below $\tau_{\text{pre}}=0.15$, so $i \in \Lambda$, $a_i=\text{DS}$, $W_i=0$, $X_i=0$, and the first case of (11) sets $d_i=0$: the kernel, the connector, and the decoder all skip it. Illegal text scores 0.22. That clears τ_{pre} , so $a_i=\text{PF}$ and $W_i=\ell_i$. A probe of that residual that is still illegal scores 0.16, below $\tau_{\text{dec}}=0.45$, and (11) stops at t_0 rather than continuing to D . Live text scores 1.00 on the residual and 0.92 once an answer marker appears, so it is prefilled and decoded to the budget. Index 0 is the second case of (11) even when its own score

is low, which is why the spine cannot be probed away. Raising τ_{pre} to 0.45 puts the illegal residual (0.22) on the same side of the cutoff as the loop. (12) then fails, F_i fires, and the residual takes EA or, under (10), never enters the kernel at all. On the GSM8K slice that is the difference between admitting 12 residuals (3848 prefill tokens) and admitting 8 (2768).

Three quantities inside one latency. Theorem 1 separates terms that a single end-to-end time adds together. Aliasing removes the k copies of L from W_{APC} and leaves one trunk, $W_{\text{CoW}} = L + \sum_i \ell_i$. The action then drops every $i \in \Lambda$ and, on EA, replaces ℓ_i by $\lceil \alpha \ell_i \rceil$, which is why $W_{\text{APP}} \leq \sum_i \ell_i$ once the trunk has been counted once. The connector is a further subset. (6) moves a page only when the residual is in $\mathcal{K}$ and the action is HP or PF; a kept hash hit still has $X_i=0$, because that page is already on the decoder. Decode length is independent of all three. ESC, Speculative Rejection, and DPTS rewrite d_i after W_i has been issued (Theorem 3), so they change D^+ and not the prefill column. A learned early-stop token can replace $\text{Score}_{\text{dec}}$ inside (11) [61]. It does not replace $\text{Score}_{\text{pre}}$, which reads ρ_i before generation. A semantic merger such as EquivPruner can shrink the residual set the harness proposes [60]; PD-Prune runs after that set exists.

The trunk is an alias. fork copies two words, the parent page table and the alias length, and increments the reference count on the trunk frames. That copy is $O(1)$ in L , which is why L is absent from (5) and from (6). A write that finds a shared page copies only the modified rows. Abort frees the residual frames of the dropped child, $b\ell_i$ bytes, and leaves the spine $M_{\text{spine}} = b(L + \ell_*)$. The same pool therefore admits

$$C = \lfloor H/M_{\text{spine}} \rfloor \quad (15)$$

sessions where a prefix cache that still holds aborted residuals would have admitted $\lfloor H/M_{\text{APC}} \rfloor$. On the measured DeepSeek-R1 GSM8K peaks, an 8 GiB pool holds 76 APC trees and 103 spines (§C). Committed decode takes the chunk budget first ((25)); whatever remains prefills the next miss tail ((26)). Speculative tokens stay outside the sampler.

Figure 2 is that alias across depth. Each edge allocates a residual and points at the parent’s pages. A leaf below the threshold frees only its own resid-

ual. A fork whose children have all died cascades back to the first ancestor that still has a live child.

4 Experiments

The comparisons below are the ones that fix the default gate. §C is the control-plane model, the per-host token ablation, the cross-model forest, and the branching sweeps.

Setup. GPU runs use vLLM on four A100-SXM4 40GB GPUs, bfloat16 weights, temperature 0, and prefix caching on. The prefill gate is measured on Qwen3-8B at $tp=1$: GSM8K with $n=16$ and $D=512$, and the first 32 MATH-500 problems at level ≥ 4 with $D=1024$. The KV comparison is the full GSM8K test ($n=1319$) on Qwen3-14B at $tp=2$. A math answer counts when the final number matches the reference. Any-finished asks that some full-budget branch match; the winner column in §C.3 asks that the committed spine match.

Prefill is host-invariant. Table 3 is Theorem 3 on the GPU. ESC, Speculative Rejection, and DPTS alone each prefill 5008 tokens, because each score is computed from tokens that already reside. They differ only in D^+ : a 512-token window, a 256-token partial-reward horizon, and a 100-token mini-step, which is why the times are 17.05 s, 13.14 s, and 11.86 s. Placing $\tau_{pre}=0.15$ in front of any of them admits 12 of 16 residuals and cuts the prefill to 3848 tokens (-23%). Any-finished stays 16/16. ESC, the host that would have decoded the loop to the end of its window, falls from 17.05 s to 14.14 s. The same gate at 0.45 admits 8 residuals, cuts the prefill to 2768 tokens, and reaches 10.93 s on ESC, still at 16/16 for this k . It is not the default. Table 4 sweeps the same ESC host across width. At $\tau_{pre}=0.15$ any-finished is 15/16, 16/16, 16/16 for $k=4, 8, 16$, and the wall-clock reduction grows with the tree (-5% , -6% , -17%) because a wider fan-out contains more repeated loops. Computed prefill falls from 1320/2624/5008 tokens to 1018/2020/3848. Raising the prefill threshold to 0.45 is faster at every width (-9% , -21% , -36%) and still 16/16 at $k=16$, but it drops illegal text together with the loop and loses two problems at $k=4$ and $k=8$. The default is the cutoff that satisfies (12) at every k .

Any-finished and the committed spine. Any-finished counts a problem when some full-budget branch matches the reference. The winner column

of Table 13 counts a problem when the committed spine matches. On this GSM8K slice the three hosts alone are all 16/16 any-finished, while the spine matches on 11/16 under ESC and on 12/16 under Speculative Rejection and DPTS. The prefill stays 5008 tokens. The hosts differ in decode tokens (131072, 98304, 78336) and in how many admitted residuals they stop later (0, 8, 8). Placing $\tau_{pre}=0.15$ in front of ESC keeps that winner at 11/16 and cuts decode tokens to 98304, because the four loops never start. The same gate in front of DPTS keeps the winner at 12/16 and finishes in 11.49 s; Speculative Rejection with the same gate finishes in 12.47 s and loses one winner, to 10/16. At 0.45 every combined row falls to a 10/16 winner: the residuals a 0.45 prefill cut-off withholds include branches that would have matched the reference. Retaining one unscored child is stricter. At the same k and $\tau=0.45$ that policy prefills 414 tokens and reaches 11/16 any-finished under ESC and 9/16 under Speculative Rejection, against 16/16 when the eight residuals that clear the score are admitted (§C.4). The gate is a threshold on every residual, and index 0 is held out of it.

The probe is the decode threshold. On MATH-500 at $k=16$, (10) drops any-finished accuracy from 20/32 to 13/32. (11) with $t_0=128$ returns the count to 20/32 and cuts end-to-end time from 104.1 s to 72.2 s. The loop is the only residual that stays at $d_i=0$. Illegal text is prefilled, probed, and extended when the prefix clears 0.45, which is the separation in (12)–(13). Table 5 is the same rule at three widths. Dropping every score below τ_{dec} cuts any-finished from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$. Skipping only the loop is safe at every width: those branches finish 0 for 0. The probe length depends on k . At $k=16$, $t_0=128$ extends every illegal residual and returns 20/32 in 72.2 s, with decode tokens falling from 524288 to 393216. At $k=8$ the shorter probe is the one that holds the count (16/32 in 42.0 s, above the full fan-out). At $k=4$ neither probe recovers 12/32; decoding the single illegal residual all the way to D is the closer point (11/32 in 23.0 s). Keeping one unscored residual instead of the eight that clear 0.45 falls to 11/16 under ESC and 9/16 under Speculative Rejection (§C.4).

The spine is the live set. On Qwen3-14B the committed spine matches APC accuracy at $0.72\times$ the median peak KV (Table 9). The same ratio

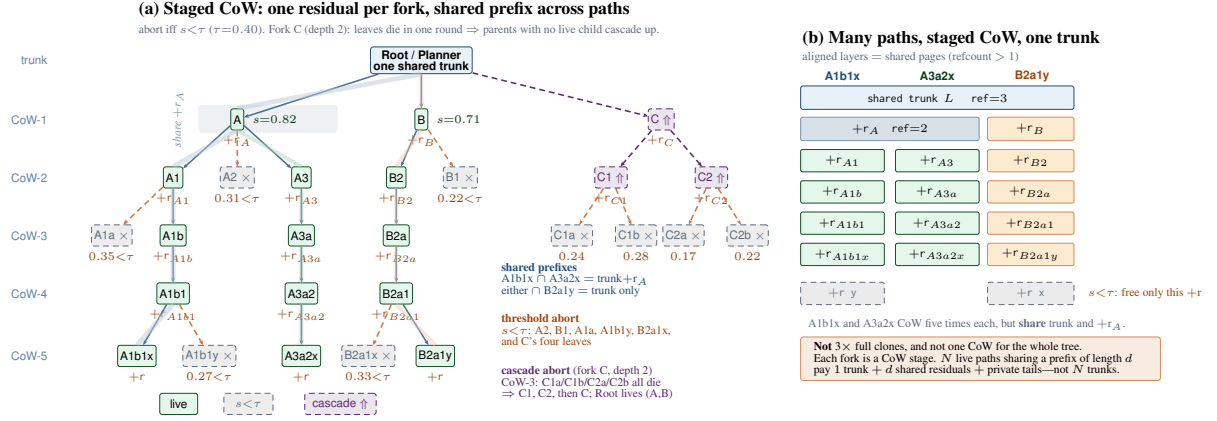


Figure 2: **Context tree.** (a) Each edge is one fork: the child aliases the parent and allocates only a residual (+r). The live paths share the trunk; A1b1x and A3a2x also share +r_A. Fork C dies at depth 2, so abort cascades to C and stops at the root, which still has A and B. (b) A shared layer is one physical page set with refcount greater than one. Aborting a leaf frees only that residual.

Table 3: Main results. The GSM8K block is Qwen3-8B, $k=16$, $D=512$. One τ_{pre} cuts prefill for every host (Theorem 3); any-finished stays 16/16. The MATH-500 block is the same model, $n=32$, $D=1024$, $\tau_{\text{dec}}=0.45$. Dropping every $s < \tau_{\text{dec}}$ before the kernel is (10). The probe is (11) at $t_0=128$. The last row is Qwen3-14B, $n=1319$, median peak KV.

		admit	prefill	time (s)	any
GSM8K. Hosts alone share the prefill; the gate moves it.					
ESC / SR / DPTS	no τ_{pre}	16	5008	17.05 / 13.14 / 11.86	16/16
ESC / SR / DPTS	$\tau_{\text{pre}}=0.15$	12	3848	14.14 / 12.47 / 11.49	16/16
ESC / SR / DPTS	$\tau_{\text{pre}}=0.45$	8	2768	10.93 / 10.80 / 10.90	16/16
MATH-500. A probe restores the branches (10) drops.					
full fan-out		16	—	104.1	20/32
drop $s < \tau_{\text{dec}}$	(10)	—	—	52.1	13/32
probe $t_0=128$	(11)	—	—	72.2	20/32
Qwen3-14B spine. $M_{\text{spine}}/M_{\text{APC}} = 0.72\times$ at accuracy 0.845 vs. 0.844.					

Table 4: GSM8K, Qwen3-8B, $n=16$, $D=512$, ESC only. Any-finished is exact match on a full-budget branch. $\tau_{\text{pre}}=0.15$ drops the loop; 0.45 also drops illegal text. Time is trunk warm plus fan-out, in seconds.

	$k=4$	$k=8$	$k=16$
ESC	16/16, 8.65	16/16, 10.84	16/16, 17.05
APP $\tau_{\text{pre}}=0.15$	15/16, 8.22	16/16, 10.19	16/16, 14.14
APP $\tau_{\text{pre}}=0.45$	14/16, 7.89	14/16, 8.60	16/16, 10.93

stays in $0.72\text{--}0.75\times$ on six model families (Figure 5). Over three Tree-of-Thoughts turns the same gate runs before each fan-out (§C.6). On Game of 24 the turns propose an equation, continue on the cards that remain, and write the expression that uses all four cards; on GSM8K they set up the algebra, rewrite the previous step, and collect the final number. Across 16 problems, three turns, and $k=4$, that is 192 residuals, of which PD-Prune+prefills 48. End-to-end time on Qwen3-8B falls from 2.48 s to 1.91 s on GSM8K and from 2.15 s to 1.52 s on Game of 24. Expand time is $7.9\text{--}10.6\times$

Table 5: MATH-500 level ≥ 4 , Qwen3-8B, $n=32$, $D=1024$. Each cell is any-finished and seconds. “Drop $s < \tau$ ” is (10). “Grow t_0 ” is (11): loops stay at $d_i=0$; every other low score is probed and continues to D only if $s'_i \geq 0.45$.

	$k=4$	$k=8$	$k=16$
Full fan-out	12/32, 25.3	15/32, 52.5	20/32, 104.1
Drop $s < \tau$	6/32, 19.1	11/32, 25.6	13/32, 52.1
Skip loops only	11/32, 23.0	17/32, 41.5	16/32, 71.4
Grow $t_0=64$	8/32, 24.0	16/32, 42.0	18/32, 72.5
Grow $t_0=128$	9/32, 24.7	13/32, 43.3	20/32, 72.2

below APC (Table 15). Accuracy stays with APC on GSM8K and rises on Game of 24, from 6/16 to 9/16 on Qwen3-4B and from 9/16 to 10/16 on Qwen3-8B (Table 6 and Figure 6).

5 Related Work

Search. Chain-of-thought writes one trace; Tree-of-Thoughts, self-consistency, and Monte Carlo tree search write many and then keep a beam, a majority, or a backed-up value [56, 45, 40, 51].

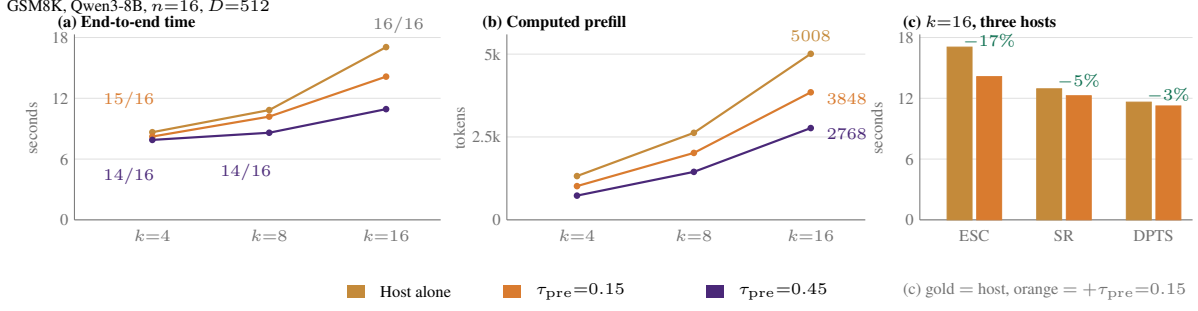


Figure 3: **Prefill threshold on GSM8K** (Tables 4 and 12). (a) Wall-clock under ESC. Labels are any-finished. $\tau_{pre}=0.45$ is 14/16 at $k=4$ and $k=8$, and 16/16 at $k=16$. (b) Computed prefill, -23% at 0.15 and -45% at 0.45. (c) The same 0.15 threshold in front of each host at $k=16$. Every row stays 16/16.



Figure 4: **Probe admission on MATH-500** (Table 5). Dropping every $s < \tau$ is the fast, low-count curve. Grow $t_0=128$ returns to 20/32 at $k=16$ and cuts 104.1 s to 72.2 s. At $k=8$, grow $t_0=64$ is 16/32 in 42.0 s, above the full fan-out.

Table 6: Three-turn Tree-of-Thoughts, $n=16$, $k=4$, $tp=2$. End-to-end time is seconds per problem. Peak KV is tokens on the live spine. FS is the alias; FS+ is PD-Prune+.

	Qwen3-4B			Qwen3-8B		
	APC	FS	FS+	APC	FS	FS+
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e (s)	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e (s)	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

The accuracy gain scales with the number of traces, and so does the token cost. EquivPruner isolates a further redundancy inside that widening: textually distinct steps can be the same mathematical action, and a learned detector keeps one representative of each equivalence class [60]. PD-Prune does not replace the search and does not decide semantic equivalence. It takes the residuals the harness has already proposed and chooses, from the residual text and the radix index, which of them are pre-filled.

Path pruning. ESC, Speculative Rejection, and DPTS drop a child after generation has started: an agreeing window, a partial-reward horizon, and a fixed mini-step [21, 31, 19]. STOP groups these signals by source and learnability and shows that a learned internal token can fire on a short prefix [61]. The shared input is text that already resides in the KV cache, so the prefill column does not move when the cut moves. Theorem 3 is that fact, and (11) is the change: a repeated loop is withheld before the kernel, and every other low score is extended past t_0 only when the probe meets τ_{dec} . A learned early-stop or equivalence signal can sit on top of that gate. The experiments keep the host as the decode policy and the residual score as the prefill policy (§C.3).

Serving and cache edits. Orca schedules at iteration granularity; vLLM pages the KV cache and hashes full blocks; SGLang shares prefixes in a radix; Sarathi overlaps a chunked prefill with decode [47, 17, 53, 2]. DistServe and Splitwise place the two phases on separate pools and move

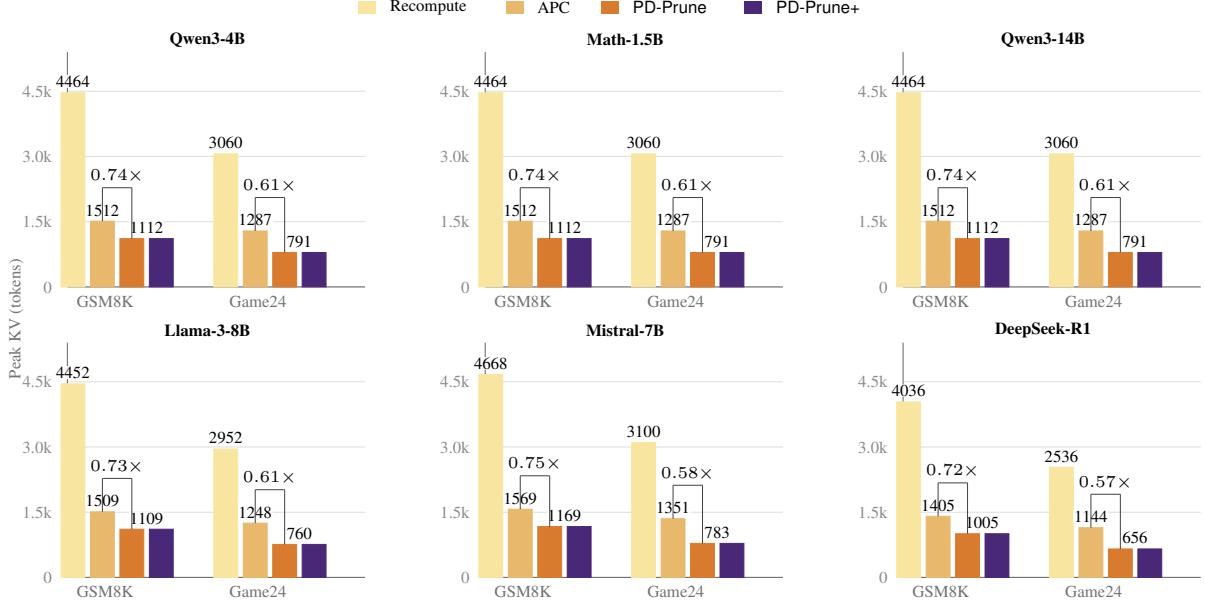


Figure 5: **Peak KV on six model families** ($n=40$, $tp=2$). Bars are recompute, APC, PD-Prune, and PD-Prune+. PD-Prune and PD-Prune+ share the spine. The annotated ratio is $M_{\text{spine}}/M_{\text{APC}}$: $0.72\text{--}0.75\times$ on GSM8K and $0.57\text{--}0.61\times$ on Game24.

every computed page across a connector [54, 32]. The hash is the right index for a replay of the winner. It is not an admission test for the children that will be aborted, which is why a same-batch fan-out still pays (2). Program schedulers raise the unit of scheduling above the request, and retention systems keep the current context across a tool gap [25, 20, 43, 38]. They leave the siblings of one fan-out unshared. CachePrune edits neurons inside one request so the model does not follow instructions embedded in retrieved context [62]; classical speculative decoding drafts output tokens rather than a future prompt prefix [18, 7]. (4) is the missing decision: one action sets W_i and X_i from the unmatched suffix. §D expands each line.

6 Conclusion

PD-Prune keeps the prefill pool and the decode pool, and changes the pages that cross between them. One action per residual sets both the kernel and the connector to the radix miss of the residuals that are kept. A repeated loop never enters that kernel. A low decode score runs a t_0 -token probe and continues only when the prefix meets τ_{dec} . The pair is (0.15, 0.45). The trunk is an alias, so the miss excludes L , and abort leaves the spine.

On Qwen3-8B the gate cuts GSM8K prefill by 23% at matched any-finished accuracy, and the MATH-500 probe restores the 20/32 a blanket drop

loses, at 72.2 s against 104.1 s. The committed spine is $0.72\text{--}0.75\times$ the prefix-cache peak.

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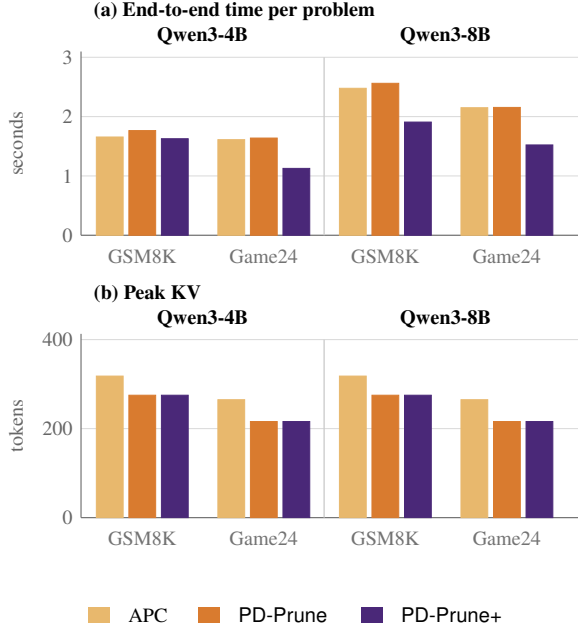


Figure 6: **Three-turn Tree-of-Thoughts** ($n=16$, $tp=2$). (a) End-to-end seconds per problem. (b) Peak KV on the live spine. PD-Prune+ prefills one of four residuals.

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A Background

Parallel reasoning and agent programs both expand one committed trunk into several residuals and then keep a single spine [45, 40, 46]. The cost of that expansion is fixed by *when* a system is allowed to reject a child. Prefix caching can share a block only after the block has been published, disaggregation moves every page the prefiller has already computed, and decode-time pruners score a child only after its prefill and a generated prefix. Table 1 records the stage. The rest of this section is why that stage falls after the tokens have been paid for, and what a fan-out looks like once the trunk, the residuals, and the spine are named separately.

A.1 Parallel Reasoning as a Fan-out

Chain-of-thought writes one trace [56]. Tree-of-Thoughts, self-consistency, and Monte Carlo tree search write many, then select [45, 40, 51]. In the Game-of-24 trace of Tree-of-Thoughts, a propose prompt first emits candidate equations and a value prompt then marks each thought sure, maybe, or impossible; breadth-first search keeps a beam of width $b=5$ [45]. The value judgment reads a thought that has already been generated. The same order appears in agent programs: a coding or tool-using harness proposes several continuations of one trunk, observes one of them, and aborts the rest [46, 44, 39]. On GSM8K and MATH the accuracy gain of that width is real [57, 58, 59], and so is the token bill. EquivPruner and early path pruning start from the same observation: most of the expanded steps do not change the answer that is finally kept [60, 61]. What they do not change is the serving order. The candidate residual is still prefilled, and under disaggregation it is still transferred, before any of those pruners runs.

A.2 Decode-Time Path Pruning

A second line drops a candidate after the model has generated it. Early-stopping self-consistency draws a window of complete samples and skips later samples only once that window agrees [21, 40]. Spec-

ulative Rejection ranks partial generations at a reward horizon and halts the lower quantile [31]. Dynamic Parallel Tree Search generates a fixed mini-step on every active node, then early-stops low-confidence children [19]. STOP organizes these signals by source and learnability and instantiates a learned internal token that fires on a generated prefix [61]. EquivPruner instead clusters semantically equivalent actions and keeps one representative of each class [60]. In every case the score is a function of text the model has already produced.

The serving consequence is visible when the three decode-time hosts are run alone on the same GSM8K fan-out (Qwen3-8B, $k=16$, $D=512$). ESC, Speculative Rejection, and DPTS each prefill 5008 tokens. They differ only after that prefill: ESC decodes to the window (131072 tokens, 17.05 s), Speculative Rejection cuts at its 256-token horizon (98304 tokens, 13.14 s), and DPTS cuts after a 100-token mini-step (78336 tokens, 11.86 s). Any-finished accuracy is 16/16 on all three. An earlier decode cut reduces the tokens emitted after the kernel. It leaves the prefill, and any connector payload computed from that prefill, unchanged. §3.2 therefore uses a second threshold, read from the residual text before the kernel, and §C.3 stacks that gate in front of the same three hosts.

A.3 Prefix Caching and Disaggregated Transfer

Automatic prefix caching and RadixAttention reuse a trunk by hashing a full block, $h = \text{hash}(\text{parent}, \text{tokens}, \text{extra})$, after those tokens exist [17, 53]. A later request hits only the blocks already in the table. A same-batch fan-out misses: the trunk has not been published, a partial block is a new block, and the engine materializes k copies of L . The hash is the right index for a replay of the winner. It is not an admission test for the children that will be aborted.

Prefill–decode disaggregation isolates time-to-first-token from inter-token latency by placing the two phases on separate instances and moving KV across a connector [54, 32]. The controller sends the request to the prefiller and then transfers the pages the prefiller computed. That path has no admission. A residual that the harness aborts after insert still occupies the connector, so the transfer equals the prefill ((2)). Program- and workflow-level schedulers raise the unit of scheduling above the request and still issue the next call as a unique continuation [25, 16, 20, 43]. Once KV retention

has removed trunk recompute, the remaining time-to-first-token term is the residual suffix of children the harness can name before the observation returns. PD-Prune reads that residual, aliases the trunk, and places on the connector only the miss tail of the residuals it keeps.

A.4 Prefill, Decode, and Idle

Autoregressive inference has two mechanically different phases [47, 54, 32]. *Prefill* consumes the prompt in parallel and is compute-bound; *decode* emits one token (or a small chunk) per step and is memory-bandwidth bound. PagedAttention [17] stores KV in non-contiguous pages; RadixAttention [53] indexes pages by token prefix so concurrent requests that share a system prompt hit a common trunk. Two consequences follow for agent workloads. First, next-turn TTFT is the prefill of tokens that miss the cache, plus queuing. Second, a memory-bound decode batch rarely saturates the tensor cores that a concurrent chunked prefill can use [2]. For agents, the prompt of turn $i+1$ is a delta on turn i , applied after a tool interval of highly variable length [20, 43].

A.5 Traditional Methods on a Fan-out

We group under *traditional methods* the mechanisms a production stack already applies to one fan-out: prefix caching, hash-indexed prefill, prefill–decode disaggregation, and decode-time pruning. Each either shares the trunk or rejects a child, and the stage of that operation is fixed. Table 1 records the stage.

First-expand cost. APC indexes a full block by $h = \text{hash}(\text{parent}, \text{tokens}, \text{extra})$ after those tokens exist [17]; RadixAttention indexes the same prefix [53]. A same-batch expansion therefore misses and clones k trunks, $W_{\text{APC}} = kL + \sum_i \ell_i$, while a later replay of $x_\star$ is a hit ($W=0$). Base-line disaggregation transfers the same span, so $X_{\text{PD}} = W_{\text{APC}}$ on the first expand ((2)). ESC [21], Speculative Rejection [31], and DPTS [19] evaluate a child only after its prefill and a generated prefix, hence after W and, under disaggregation, after X .

A.6 Fan-out Model

Definition 1 names the same objects used below. A parent u holds a committed trunk x_u with $|x_u|=L$. A fan-out is a set of residuals $\{\rho_i\}_{i=1}^k$, $|\rho_i|=\ell_i$. The bound winner $\star$ leaves the spine $x_\star = x_u \parallel \rho_\star$,

Table 7: Notation for Definition 1 and for (4)–(6).

Symbol	Domain	Meaning
$L, \ell_i, \ell_\star$	$\mathbb{N}$	trunk; residual i ; committed residual
ρ_i, x_u	tokens	residual of child i ; committed trunk
h_i	digest	$\text{hash}(\text{parent}, \rho_i[0:P], \text{extra})$
m_i	tokens	full-page prefix of ρ_i in H , steps of P
s_i, s'_i, τ	$[0, 1]$	score of ρ_i ; score of the probe y_i ; threshold
t_0, D, d_i	tokens	probe length; decode budget; length assigned to i
α	$(0, 1]$	early-abort prefix fraction (0.20)
p_i, q_i	$[0, 1]$	admit probability; residual-prefix hit rate
$a_i \in \mathcal{A}$	$\mathcal{A}$	HS, HP, DS, EA, PF ((4))
Λ, F_i	set, $\{0, 1\}$	repeated-loop index set; early-abort predicate
$\mathcal{K}$	$\subseteq [k]$	residuals that remain after admission
W_i, X_i	tokens	GPU miss tail; connector payload
b, P	bytes, tok	KV bytes per token; page size
$B_t = B_t^c + B_t^s$	tokens	committed and speculative budget on tick t

$|x_\star| = L + \ell_\star$. Let b be KV bytes per token and P the page size. The three occupancies are (1): $M_{\text{clone}} = b(kL + \sum_i \ell_i)$ is the first-expand footprint of recompute, APC, and baseline disaggregation; $M_{\text{CoW}} = b(L + \sum_i \ell_i)$ is one shared trunk with every residual live; $M_{\text{spine}} = b(L + \ell_\star)$ is the footprint after abort. W is the token count issued to a prefill kernel before $\star$ is known; X is the token count inserted on the P/D connector. Table 7 lists the remaining symbols.

B System Detail

PD-Prune lies between the agent runtime and the LLM engine (Figure 1). The harness exposes branch structure; the control plane admits, prunes, binds, and retains; the backend moves pages and tokens.

B.1 API

<code>open(session)</code>	Create a root; prefill system+user.
<code>fork(u, bid, tok?)</code>	Child of u ; optional known suffix.
<code>speculate(v, pri)</code>	Mark v eligible for speculative prefill.
<code>commit(u, tokens)</code>	Bind by LCP; start decode.
<code>join(children, pol)</code>	Merge fan-out (all/first/kofn).
<code>abort(v) / close</code>	Drop a subtree; end the session.

`commit` is the only operator on the critical path of user-visible tokens.

B.2 Copy-on-Write Context Tree

Definition 3 (Context tree). A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u such that $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|:]$, a mode $\rho_u \in \{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u , and a logical page table $\sigma_u = (\text{parent}, \text{alias_len}, \rho_u^{\text{pages}})$.

Invariants: (i) prefix; (ii) at most one committed child per node except during join; (iii) speculative nodes are invisible to sampling; (iv) a page’s refcount equals the number of non-dead nodes that map it. `fork( $u, \text{bid}, x^k$ )` copies two words ($\text{parent} \leftarrow \sigma_u$, $\text{alias_len} \leftarrow |x_u|$) and increfs the aliased frames— $O(1)$ in L . A write that hits $\text{ref} > 1$ or a read-only tail CoW-copies only the valid rows. Figure 7 is that pool. A multi-level fan-out is a *sequence* of CoWs (Figure 2): each edge aliases the parent and allocates only $+r$; live paths share every page on the common prefix; abort is residual-only, and a subtree with no live child cascades until the committed spine. The two compose: LCP commit publishes full winner pages into the APC index so a later session can hash-hit them (§3.1). Abort frees $b\ell_i$ bytes, not $b(L + \ell_i)$. TTL and offload are node-granular: speculative leaves first, then idle committed leaves, then the spine [20, 43].

B.3 Admission and Commit

A candidate of parent u is a pair $(x^k, \hat{x}^o)$ with admit probability p and prefix-hit rate q . Known suffixes ($p=q=1$) precede observation residuals ($q<1$), subject to $T_{\text{idle}}(u)$, residual HBM, and branch cap m . Admission stores a future prompt prefix rather than draft output tokens [18, 7].

LCP commit. On prompt x at u ,

$$v^\star = \arg \max_v \text{LCP}(x, x_v), \quad g_u \leftarrow g_u + 1. \quad (16)$$

The main-text statement is (8). Pages of the LCP become committed; the unmatched tail is a committed prefill; every sibling aborts. A job tagged $g \neq g_u$ drops in $O(1)$. Committed decode attends only $\text{KV}(x_{v^\star})$, so the emitted tokens match a baseline that prefills committed tokens only.

B.4 Admission Rule

One action a_i fixes the pair (W_i, X_i) for residual ρ_i (Figure 8). a_i reads m_i first, because a radix match costs no model FLOPs. On a miss it reads $i \in \Lambda$, because a repeated loop is known before a kernel. Only a miss outside Λ enters the kernel, and F_i stops that kernel at $\lceil \alpha \ell_i \rceil$. X_i equals W_i on $\mathcal{K}$ and is zero otherwise, so the connector moves the unmatched suffix of retained residuals. Index 0 is in $\mathcal{K}$; pages stay out of H until (8). The alias of §B.2 is what removes L from both quantities. On Qwen3-14B GSM8K the alias already stores

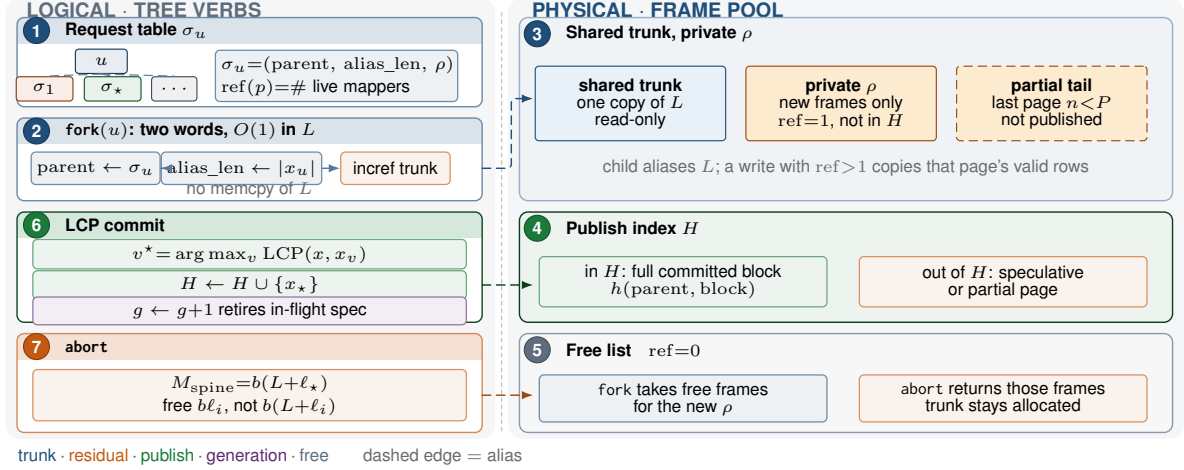


Figure 7: **Context-tree page pool.** fork aliases the trunk in $O(1)$ and allocates only ρ ①–③. LCP commit publishes x_* into H ④⑥. abort returns residual frames only, so live occupancy is the spine $M_{\text{spine}}=b(L+\ell_*)$ ⑤⑦.

less KV than APC, while median fan-out is 140 ms against 87 ms because every residual is still pre-filled; a_i is the term that removes that prefill.

Figure 2 draws the same staged alias: shared prefixes stay one physical page set, and a dead fork returns only the residuals along the cascade.

Action. Let H be the published radix index and P the page size. The matched prefix, the loop set, and the early-abort bit are

$$m_i = \max\{jP : jP \leq \ell_i, \rho_i[0:jP] \in H\}, \quad (17)$$

$$\Lambda = \{i \neq 0 : \text{Loop}(\rho_i)\},$$

$$F_i = [s(\rho_i[0:\lceil \alpha \ell_i \rceil]) < \tau].$$

The trunk is the alias $\sigma_i \leftarrow (\sigma_u, |x_u|)$, so $L \notin W_i$. With $\mathcal{A} = \{\text{HS}, \text{HP}, \text{DS}, \text{EA}, \text{PF}\}$ and first-match Iverson brackets,

$$a_i = \text{HS}[m_i=\ell_i] + \text{HP}[0 < m_i < \ell_i] \\ + \text{DS}[m_i=0][i \in \Lambda] + \text{EA}[m_i=0][i \notin \Lambda][F_i] \\ + \text{PF}[m_i=0][i \notin \Lambda][\neg F_i]. \quad (18)$$

Index 0 is forced into $\mathcal{K}$, and $a_0 \in \{\text{HS}, \text{HP}, \text{PF}\}$.

$$W_i = \lceil \alpha \ell_i \rceil [a_i=\text{EA}] + (\ell_i - m_i)[a_i=\text{HP}] \\ + \ell_i[a_i=\text{PF}], \quad (19)$$

$$X_i = W_i[i \in \mathcal{K}][a_i \in \{\text{HP}, \text{PF}\}]. \quad (20)$$

Figure 10 is (4)–(6) on one radix tree: green pages lie in H , orange pages are grown, gray pages are withheld.

$$d_i^\times = D \cdot [i = 0 \vee s_i \geq \tau]. \quad (21)$$

Algorithm 2 Cold-index restriction of Algorithm 1: every $m_i=0$. Probe admission, replacing (10). Λ is the repeated-loop set. y_i is a probe of t_0 tokens and s'_i is its draft score. Iverson brackets select the decode length. DS applies only to Λ .

Input: $\{\rho_i\}_{i=0}^{k-1}$, threshold τ , probe t_0 , budget D
Output: action a_i , decode length d_i , set $\mathcal{K}$

- 1: $\Lambda \leftarrow \{i \neq 0 : \text{Loop}(\rho_i)\}$
- 2: **for** $i = 0$ **to** $k - 1$ **do**
- 3: $s_i \leftarrow \text{DraftScore}(\rho_i)$
- 4: **if** $i \in \Lambda$ **then**
- 5: $(a_i, d_i) \leftarrow (\text{DS}, 0) \{(\text{10}) \text{ set } d_i^\times=0 \text{ for all } s_i < \tau\}$
- 6: **else if** $i = 0 \vee s_i \geq \tau$ **then**
- 7: $(a_i, d_i) \leftarrow (\text{PF}, D)$
- 8: **else**
- 9: $y_i \leftarrow \text{Dec}(x_u \parallel \rho_i, t_0)$
- 10: $s'_i \leftarrow \text{DraftScore}(y_i)$
- 11: $(a_i, d_i) \leftarrow (\text{PF}, D[s'_i \geq \tau] + t_0[s'_i < \tau])$
- 12: **end if**
- 13: **end for**
- 14: $\mathcal{K} \leftarrow \{i : a_i \neq \text{DS}\}$
- 15: $W_i \leftarrow \ell_i \cdot [a_i = \text{PF}]$
- 16: $\text{DecTokens} \leftarrow \sum_{i=0}^{k-1} d_i$

(10) drops every $s_i < \tau$ before prefill. Algorithm 1 keeps DS only for $i \in \Lambda$; every other low score is prefilled and probed. Table 8 is the resulting first-expand span.

Theorem 1 is this comparison: on a cold index, $W_{\text{APP}} \leq W_{\text{CoW}} \leq W_{\text{APC}} = W_{\text{PD}}$ and $X_{\text{APP}} \leq X_{\text{PD}}$, strictly in W whenever a repeated loop is withheld, and strictly in X whenever $\mathcal{K} \neq [k]$ or any kept residual is a hash hit.

From the action to the wire. The index key is hash(parent, block_tokens, extra) [17], and only full pages enter H , so m_i in (3) steps by P . (8) is what later sets $m_i=\ell_i$. A hit is already visible

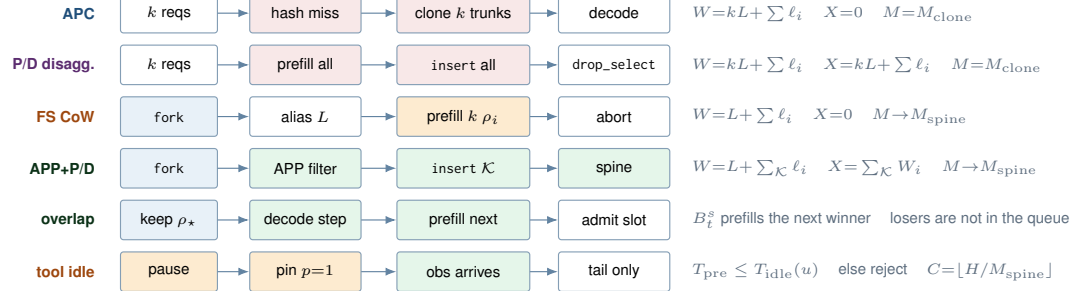
ρ_i : i -th residual h_i : block hash s_i : draft score α : early-abort fraction κ : chunk W_i : GPU miss tail X_i : connector tokens a_i : action

(a) APP + P/D pipeline (Alg. 1)



Parallelism: k residuals Layout: location-addressed CoW, not content-addressed APC Winner $i=0$ is never dropped Spec pages unpublished
Work $W = \sum_i W_i$ Transfer $X = \sum_{i \in \mathcal{K}} W_i$ Occupancy after abort $M_{\text{spine}} = b(L + \ell_*)$ Replay of x_* : $W = X = 0$

(b) First-fan-out dataflow (same k, L, ℓ). Bottleneck in red.



Takeaway: pruning removes non-winner work; the spine then sets the slot count, shares the decode step with the next winner, and absorbs a known suffix during the tool pause.

Figure 8: **The admission rule and where it sits on the path.** (a) a_i is the first match of (4). Only W_i for $i \in \mathcal{K}$ reaches insert. (b) APC and baseline disaggregation copy the trunk. The alias removes L from W and from X . The overlap row prefills the next miss during decode. The tool-interval row prefills a suffix with $p=1$ while $T_{\text{pre}} \leq T_{\text{idle}}$.

Table 8: Work, transfer, and memory spans (bottleneck in red). W_{pre} is GPU prefill tokens; X is connector tokens; M is live KV after abort; T_{ab} is abort cost in pages; replay is a later open of the published winner. PAR is the number of trunk copies touched on the first expand.

	W_{pre}	X	M after abort	T_{ab}	PAR _{trunk}	Replay W_{pre}
Recompute	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	$\Theta(kL/P)$	k	$kL + \ell_*$
APC (first expand)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU of hashed ρ	k	0
hash prefill (first)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU	k	0
P/D disagg.	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	full KV transfer	k	0
FS CoW	$L + \sum_i \ell_i$	0	$L + \ell_*$	$\Theta(\ell_i/P)$	1	L_{miss}
FS+ + P/D	$L + \sum_{i \in \mathcal{K}} \ell_i$	$\sum_{i \in \mathcal{K}} W_i$	$L + \ell_*$	lazy, residual only	1	0

$P=16$. Known-suffix items ($p=1$) are admitted before residuals ($p < 1$). Spec pages are never hashed; LCP commit publishes full blocks of x_* .

on the decoder, hence $X_i=0$ when $a_i=\text{HS}$. (10) withholds every $s_i < \tau$; Algorithm 1 restricts that withhold to $i \in \Lambda$ and continues any other low score past t_0 iff $s'_i \geq \tau$. The t_0 probe is a decode prefix; F_i is a prefill prefix. Defaults are $\tau_{\text{pre}}=0.15$, and on MATH-500 $\tau=0.45$, $t_0 \in \{64, 128\}$. Baseline disaggregation moves (2); the rule moves (6).

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{CoW}} + T_{\text{abort}}. \quad (22)$$

(9) is the same identity. T_{CoW} is a pointer swap. §C.2 evaluates (10) at 139 ms (APC), 162 ms (baseline disaggregation), 47 ms (alias only), and 31.4 ms. §C.2 uses Algorithm 1 and cuts computed prefill by 23%.

B.5 Separate Prefill and Decode Thresholds

The prefill score and the decode score read different text, so PD-Prune stores them as a pair $(\tau_{\text{pre}}, \tau_{\text{dec}}) = (0.15, 0.45)$ rather than one threshold reused at both stages. τ_{pre} is applied to the

residual before the kernel and decides whether W_i is issued. τ_{dec} is applied to the generated probe and decides whether that probe continues from t_0 to the budget D . Algorithm 1 already separates the two decisions: a repeated loop takes DS and never enters the kernel; every other residual is prefilled; a residual whose draft score is below τ_{dec} is extended only when the probe score s'_i meets τ_{dec} . Index 0 stays in $\mathcal{K}$, so a low score cannot drop the spine.

Table 2 is why the two numbers are not interchangeable. On the residual, a loop scores about 0.02 and illegal text about 0.22, so $\tau_{\text{pre}}=0.15$ withholds the loop and still prefills the illegal residual. On the generated prefix, text that is still illegal scores about 0.16, while a prefix that has written ##### or a closed \boxed scores about 0.92. The decode threshold therefore has to sit above 0.16 and below the finished marker. Using 0.45 as the prefill threshold drops the illegal residual together

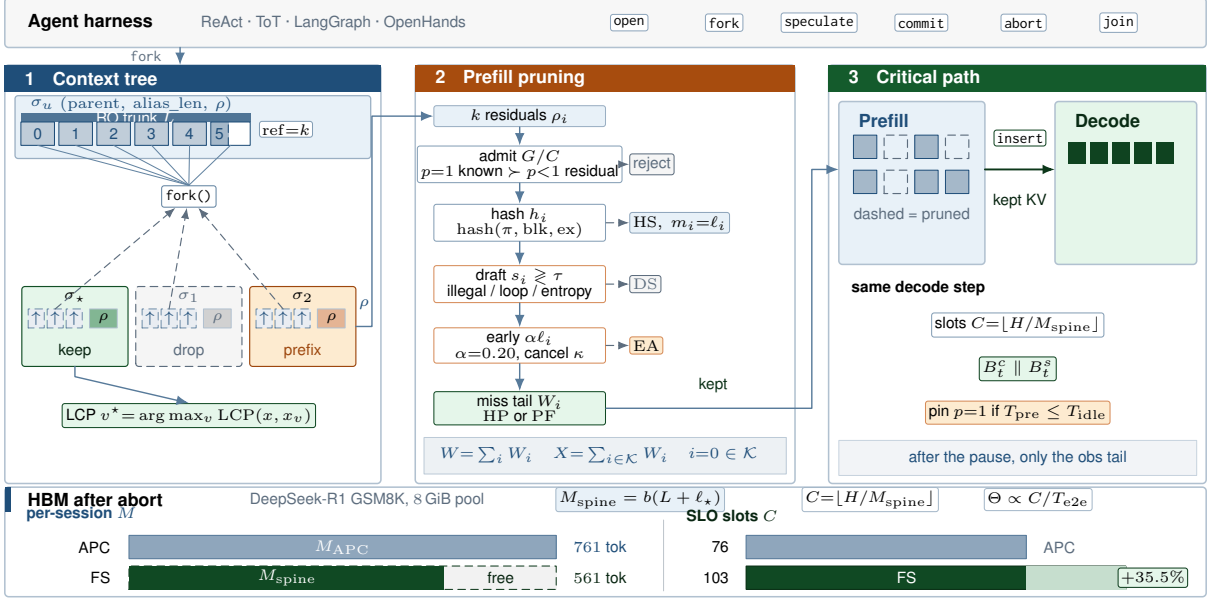


Figure 9: **Control plane behind Figure 1.** (1) σ_u aliases one trunk; fork is $O(1)$; (8) binds the winner. (2) One action a_i ((4)). A full-page hit issues no kernel; a partial hit prefills the miss; $i \in \Lambda$ is withheld. X sums W_i over $\mathcal{K}$. (3) After abort the live set is the spine.

with the loop, which is the blanket withhold of (10). Using 0.15 as the decode threshold continues a probe that is still illegal, because 0.16 remains above the line, and the decode runs out to D . The probe is the step between them. Loops stop before the kernel. Every other residual below τ_{dec} is pre-filled, decoded for t_0 tokens, and extended to D only when $s'_i \geq \tau_{\text{dec}}$. §C.2 measures that split on MATH-500: withholding every score below 0.45 drops any-finished accuracy from 20/32 to 13/32 at $k=16$, and a 128-token probe restores 20/32.

B.6 Critical-Path Placement

Abort under the rule leaves $M_{\text{spine}} = b(L + \ell_*)$, so the same pool admits

$$C = \lfloor H/M \rfloor \quad (23)$$

sessions, with $M = M_{\text{spine}}$ for PD-Prune and $M = M_{\text{APC}}$ while a prefix cache still holds aborted residuals.

$$\Theta \propto C/T_{e2e}. \quad (24)$$

On tick t the chunk budget [2] splits by the committed rate $\underline{s}_r$ [52],

$$B_t^c = \min\left(B_t, \sum_{r \in R_t^c} \max(\underline{s}_r \Delta, 1)\right), \quad (25)$$

$$B_t^s = B_t - B_t^c. \quad (26)$$

B_t^c decodes the spine. B_t^s prefills the next miss tail, or a known suffix ($p=1$) when $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$. The observation then pays only the unmatched tail under (8). Speculative tokens stay outside the sampler.

C Full Evaluation

The GPU measurements share one node and one engine, and they differ by model, tensor-parallel width, and workload. §C.1 is the full GSM8K forest on Qwen3-14B at $\text{tp}=2$. §C.2 is the earlier blanket-skip cost model of APP, with no GPU, against APC, hash prefill, baseline prefill–decode disaggregation, PD-Prune, and three decoding-time pruners; §C.2 supersedes that predicate. §C.2 measures the GSM8K prefill threshold $\tau_{\text{pre}}=0.15$ on Qwen3-8B and the accuracy–latency trade-off when that threshold precedes ESC, Speculative Rejection, and DPTS. §C.5 runs PD-Prune+on Qwen3-8B at $\text{tp}=2$ and $\text{tp}=4$. §C.6 is a three-turn Tree-of-Thoughts on Qwen3-4B and Qwen3-8B, one problem per generate. §C.7 repeats the forest on six families at $n=40$. §C.8 sweeps branching, depth, pruning, and batch width on Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B, then runs a 2048-token contest forest on that DeepSeek model and on Qwen3-4B.

Setup. GPU runs use vLLM on four A100-SXM4 40GB GPUs, with bfloat16 weights and

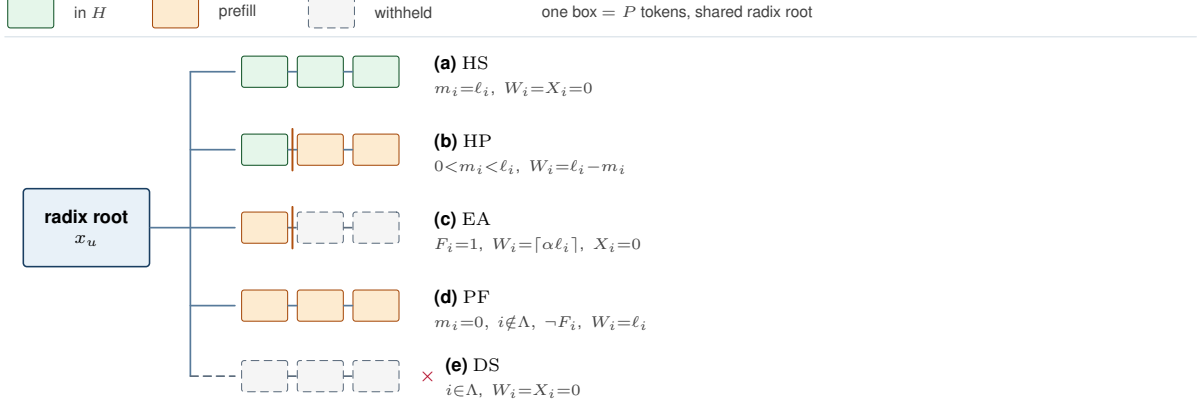


Figure 10: **One radix tree, five actions** ((4)–(6)). The root is the aliased trunk x_u . Each edge is one page of P tokens. Green pages are in H ; orange pages are prefilled; gray pages are withheld. (a) HS: $m_i = \ell_i$, $W_i = X_i = 0$. (b) HP: $0 < m_i < \ell_i$, $W_i = \ell_i - m_i$. (c) EA: $m_i = 0$, $F_i = 1$, $W_i = \lceil \alpha \ell_i \rceil$, $X_i = 0$. (d) PF: $m_i = 0$, $i \notin \Lambda$, $\neg F_i$, $W_i = \ell_i$. (e) DS: $i \in \Lambda$, $W_i = X_i = 0$. A score below τ with $i \notin \Lambda$ is prefilled and probed (Algorithm 1).

Table 9: GSM8K test, Qwen3-14B, $\text{tp}=2$, $n=1319$. Median over 165 batches of eight. Best storage in green.

	recompute	APC	FS
Accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	$2.89 \times$	$1.00 \times$	$0.72 \times$
Fan-out, median (ms)	326	87	140
KV saving vs. rec.	—	0.65	0.75

$\text{tp}=2$ or 4. The models are Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B. Tree-of-Thoughts uses $k=4$ and a 512-token budget (2048 on the contest forest), in batches of eight. The three-turn run in §C.6 issues one problem per generate. A math answer is correct when its final number matches the reference.

C.1 GPU forest

Setup. This section uses Qwen3-14B at $\text{tp}=2$. The GSM8K test set has 1319 problems. Each problem is expanded with four thoughts and decoded for at most 512 tokens, in 165 batches of eight. Accuracy is the fraction of problems whose final number matches the reference.

Storage and expand latency. Table 9 reports the storage result and the expand-latency gap that APP is designed to close. Median peak KV is $0.72 \times$ APC and $0.25 \times$ recompute, and accuracy stays within one point of both. Storage and fan-out do not share that ordering: median fan-out is 140 ms for PD-Prune and 87 ms for APC, because copy-on-write and abort execute on the expand path, whereas an APC hash hit does not. End-to-

end time falls once the prefill rule runs. On Qwen3-8B MATH-500 at $k=16$, a 128-token probe keeps any-finished at 20/32 and cuts end-to-end time from 104.1 s to 72.2 s (Table 5). On GSM8K at the same width, $\tau_{\text{pre}}=0.15$ cuts ESC from 17.05 s to 14.14 s at 16/16, and $\tau_{\text{pre}}=0.45$ reaches 10.93 s at the same accuracy (Table 4). The same peak ordering, $0.58\text{--}0.80 \times$ APC, holds on seven math sets for Qwen3-14B, Qwen3-8B, and DeepSeek-R1-Distill-Llama-8B at $\text{tp}=2$ and $\text{tp}=4$. The peak is unchanged when the tensor width doubles; it follows the committed path, as (1) requires.

C.2 Prefill pruning

Setup. This subsection reports the earlier blanket-skip cost model. The GPU admission rule is §C.2 and §C.2. Control plane, no GPU. Ten sessions, $k=4$, $L=256$, $\ell=32$. Two of four residuals fail the draft score and are withheld before prefill; the winner is index 0. Prefill time is $\ell \times 12 \mu\text{s}/\text{token}$; P/D transfer is $2 \mu\text{s}/\text{token}$. APC and hash prefill publish blocks only after a request materializes them, so the first fan-out clones k trunks. Baseline disaggregation uses that prefill and then transfers the live token span. PD-Prune aliases the trunk and prefills the k residuals. PD-Prune+ applies the residual action under the stricter predicate (10): every $s_i < \tau$ is withheld before the kernel, and the connector carries only $\mathcal{K}$.

Fan-out. Table 10 is the comparison motivated by the GPU latency gap. Figure 11 shows occupancy over the expand–abort–decode phase. On the first expand, hash prefill matches APC: the blocks are unpublished, so the hash probe does not re-

Table 10: Control-plane fan-out (10 sessions, $k=4$, $L=256$, $\ell=32$). Bottleneck in red; PD-Prune+in green. Not GPU wall-clock.

Phase	Method	W	T_{fan}	X	peak	skip	prune
fan-out	APC	11520	139.0	0	11520	0	0
	hash	11520	139.0	0	11520	0	0
	P/D disagg.	11520	162.1	11520	11520	0	0
	FS	3840	47.1	0	3840	0	0
	PD-Prune+	2592	31.4	32	2880	9	30
PD-Prune+/ APC		-77.5%	-77.4%	—	-75%	—	—

Table 11: Decoding-time pruning on the Table 10 fan-out (10×4 , $L=256$). Decode-to-cut is the prefix that must be generated before a child can be dropped. “On losers” counts that prefix on the two low-score children. Same 12 μs /token clock. Not GPU wall-clock.

	prefill	decode	on losers	peak	T_{cut}
ESC	2560	20480	10240	23040	276.5
Spec. Rejection	2560	10240	5120	12800	153.6
DPTS	2560	4000	2000	6560	78.7
PD-Prune+	2592	0	0	2880	31.4

move the cloned trunks. Baseline disaggregation exceeds APC (162 vs. 139 ms) because it transfers that clone. PD-Prune removes the clone by aliasing the trunk (3840 prefill tokens, 47 ms). APP then withholds prefill from 30 of the 40 residuals, which reduces T_{fan} to 31.4 ms and the connector payload from 11520 tokens to 32.

Decoding-time pruning. Table 11 runs ESC [21], Speculative Rejection [31], and DPTS [19] on this same fan-out. Each method prefills the shared trunk once. The length of the decode that produces the prune score is the length published for that method. ESC stops after a window of 5 agreeing samples; this fan-out has four children and two low-score answers, so the window stays open and every child is decoded to the 512-token budget. Speculative Rejection is charged its partial-reward horizon $\tau=256$ and then drops the lower half ($\alpha=0.5$); the memory-limit trigger in that paper would not fire at $k=4$, so τ is the earlier checkpoint. DPTS early-stops after its 100-token mini-step. The two low-score children are the ones removed, and the winner is kept. Decoded tokens use the same 12 μs /token clock. DPTS is the earliest of the three and still decodes 4000 tokens before the drop, 2000 of them on the rejected children, reaching the cut at 78.7 ms. Under this cost model APP has already withheld those residuals: 0 tokens on rejected children and $T_{\text{fan}}=31.4$ ms. That row is the blanket skip of (10). §C.2 supersedes it: only a repeated loop is withheld before prefill.

Probe admission. (10) is inadmissible once the residual is not a loop. We measure Algorithm 1 on Qwen3-8B, one GPU each, four A100-40GB, $\text{tp}=1$. The slice is the first 32 MATH-500 problems at level 4 or 5. Temperature is 0, EOS is ignored, prefix caching is on, and the decode budget is $D=1024$. Half of the k residuals are contest strategies. The other half alternates a repeated loop with illegal text. A problem counts when some branch that ran all 1024 tokens matches the reference answer. The winner is residual 0. Table 5 is the correction. Dropping every residual below τ cuts any-finished from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$. Loop branches are 0 for 0 at every k , so skipping them before prefill is safe. At $k=16$, a 128-token probe extends every illegal residual (128/128) and none of the loops. Any-finished returns to the full fan-out, 20/32, while end-to-end time falls from 104.1 s to 72.2 s and decode tokens from 524288 to 393216. Probing the loops as well extends none of them and costs 74.4 s for the same 20/32. At $k=8$ the shorter probe is the one that holds accuracy: grow at $P=64$ is 16/32 in 42.0 s, against 15/32 in 52.5 s for the full fan-out. At $k=4$ the probe does not recover the full fan-out (9/32 against 12/32); skipping the single loop, and decoding the illegal residual to D , is 11/32 in 23.0 s. Figure 4 is Table 5: at $k=16$ grow $t_0=128$ meets the full-fan-out count at 72.2 s against 104.1 s.

Accuracy–speed trade-off. The MATH-500 probe still uses a decode threshold $\tau=0.45$ on the generated prefix. Prefill admission is a different score, with default $\tau_{\text{pre}}=0.15$: it drops only the repeated loop (0.02) and starts illegal text (0.22) and every live strategy (1.0). We measure that rule on Qwen3-8B GSM8K, $n=16$, $D=512$, the same hopeless mix, four A100-40GB, $\text{tp}=1$. The host decoder is ESC, so a withheld child never pays residual prefill or a decode slot. A denser sweep of τ_{pre} collapses to three admission sets; 0.15 and 0.45 are the two nonzero operating points. Table 4 is the accuracy–latency trade-off. $\tau_{\text{pre}}=0.15$ attains 15/16, 16/16, 16/16 and reduces wall-clock time as the tree widens (−5%, −6%, −17%). Computed prefill tokens fall from 1320/2624/5008 to 1018/2020/3848. Raising τ_{pre} to 0.45 reduces end-to-end time by −9%, −21%, −36%, with 732/1448/2768 computed prefill tokens. Accuracy falls by two problems at $k=4$ and $k=8$. At $k=16$ both thresholds remain at 16/16, so 0.45

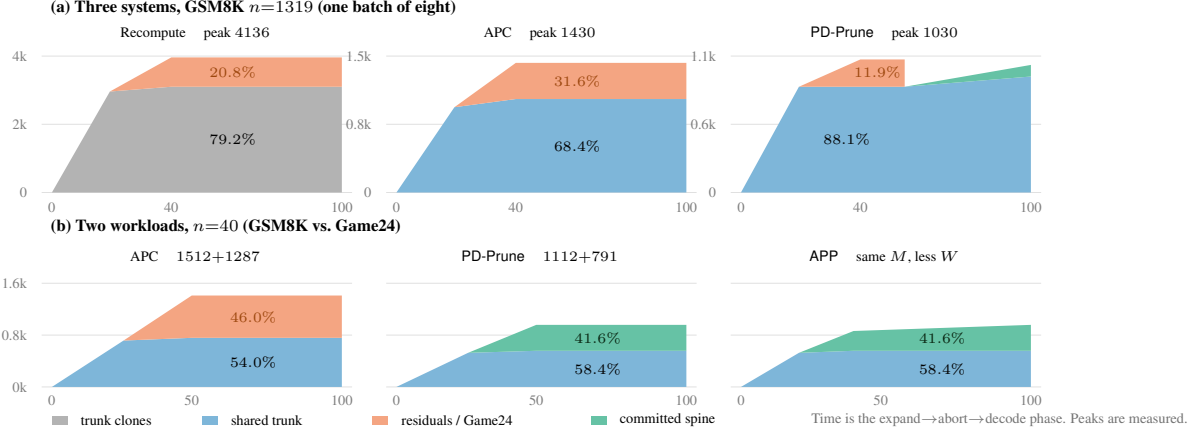


Figure 11: **Stacked KV occupancy** over the expand→abort→decode phase. (a) One GSM8K batch of eight on Qwen3-14B (per-panel y -scale; peaks 4136 / 1430 / 1030). Annotated fractions are trunk vs. residual under the $k=4$, $L=256$, $\ell=32$ model. (b) Two workloads ($n=40$): GSM8K (bottom) plus Game24 (top) for APC, PD-Prune, and APP. PD-Prune and APP share M ; APP drops residual W earlier. Percentages are the share of the stacked live tokens.

Table 12: Same GSM8K slice, $k=16$, $\tau_{\text{pre}}=0.15$ in front of three host decoders. Admitted children and computed prefill tokens are identical across hosts: prefill admission does not read τ , d_{DPTS} , or α . Any-finished stays 16/16.

	adm.	prefill	e2e (s)	any
ESC	16	5008	17.05	16/16
ESC+APP	12	3848	14.14	16/16
Spec. Rejection	16	5008	13.14	16/16
SR+APP	12	3848	12.47	16/16
DPTS	16	5008	11.86	16/16
DPTS+APP	12	3848	11.49	16/16

is the faster matched-accuracy setting (10.93 s vs. 14.14 s). The default is 0.15 because accuracy holds at every k .

Table 12 stacks the same $\tau_{\text{pre}}=0.15$ in front of each host. ESC, Speculative Rejection, and DPTS alone all report 5008 computed prefill tokens: a decode-time cut still prefills every child. Adding APP admits 12 of 16 residuals and 3848 prefill tokens on every host. Wall-clock falls 17% on ESC, 5% on Speculative Rejection, and 3% on DPTS, because those last two hosts would have stopped the loop later anyway. The accuracy that Table 4 preserves is unchanged: every row in Table 12 is 16/16. At $k=8$ the same threshold is 16/16 on ESC (10.84 to 10.19 s) and lifts Speculative Rejection and DPTS from 15/16 to 16/16. Figure 3 plots these rows. Prefill admission is identical across the three hosts: 16 children and 5008 tokens become 12 and 3848.

C.3 Decoding, prefill, and the combination

Table 12 places one prefill threshold in front of three hosts. Table 13 separates the three ways those hosts can be combined with the gate, on the same Qwen3-8B GSM8K slice at $k=16$. Any-finished accuracy is 16/16 on every row. The winner column is the number of problems on which the committed spine, rather than some other full-budget branch, matches the reference.

Decoding pruning leaves the prefill at 5008 tokens because the host reads a generated prefix. The hosts differ in where that prefix ends. ESC runs each child to the 512-token budget, Speculative Rejection stops the lower half at a 256-token horizon, and DPTS stops after a 100-token mini-step. Wall-clock falls from 17.05 s to 13.14 s to 11.86 s, and the winner count moves from 11/16 to 12/16, while the prefill column does not move. Prefill pruning alone admits 12 residuals at $\tau_{\text{pre}}=0.15$ and 8 at 0.45, cuts computed prefill to 3848 and 2768 tokens, and then decodes every admitted residual to D . No residual is stopped later, which is why the 0.15 row still decodes 98304 tokens and the 0.45 row decodes 65536. The winner stays at 11/16 for 0.15, matching ESC, and falls by one problem at 0.45.

The combined rows keep the prefill reduction and add the host cut. Speculative Rejection with $\tau_{\text{pre}}=0.15$ admits 12 and stops 4 more at the horizon, finishing in 12.47 s; DPTS with the same gate finishes in 11.49 s with the winner still at 12/16. At $\tau_{\text{pre}}=0.45$ the decode-side drop count is zero:

Table 13: GSM8K, Qwen3-8B, $n=16$, $k=16$, $D=512$. Decoding pruning sets no τ_{pre} and cuts only during decode. Prefill pruning cuts before the kernel and then decodes every admitted residual to D . The combined rows apply the same τ_{pre} and then the host’s own cut. “Stopped later” is the number of admitted residuals the host drops at its decode cut. Any-finished stays 16/16.

Setting	admit	stopped later	prefill	decode	e2e (s)	winner
<i>Decoding pruning: no τ_{pre}</i>						
ESC, budget $D=512$	16	0	5008	131072	17.05	11/16
Spec. Rejection, horizon 256	16	8	5008	98304	13.14	12/16
DPTS, mini-step 100	16	8	5008	78336	11.86	12/16
<i>Prefill pruning: decode runs to D</i>						
$\tau_{\text{pre}}=0.15$	12	0	3848	98304	14.14	11/16
$\tau_{\text{pre}}=0.45$	8	0	2768	65536	10.93	10/16
<i>Combined: τ_{pre} in front of the host cut</i>						
SR + $\tau_{\text{pre}}=0.15$	12	4	3848	81920	12.47	10/16
DPTS + $\tau_{\text{pre}}=0.15$	12	4	3848	71936	11.49	12/16
SR + $\tau_{\text{pre}}=0.45$	8	0	2768	65536	10.80	10/16
DPTS + $\tau_{\text{pre}}=0.45$	8	0	2768	65536	10.90	10/16

the residuals the host would have cut were already withheld, both hosts decode 65536 tokens, and end-to-end time is 10.80 s and 10.90 s. Any-finished accuracy remains 16/16. The default gate is 0.15 because, as Table 4 shows, 0.45 loses two problems at $k=4$ and $k=8$.

C.4 Score-based admission

Keeping only the single best residual is a different rule from scoring every residual against τ . On the same GSM8K slice at $k=16$ and $\tau=0.45$, an admission policy that ignores the draft score and retains one child prefills 414 tokens and reaches 11/16 any-finished under ESC and 9/16 under Speculative Rejection. Scoring the residuals and admitting the eight that clear τ prefills 2768 tokens and restores 16/16 on both hosts. Index 0 is always kept; the score is what decides which of the other children are worth a kernel. A one-residual spine removes correct branches that a 0.45 threshold would still have probed.

Concurrency from measured KV. Our empirical results on DeepSeek-R1 GSM8K (761 APC tokens, 561 PD-Prune tokens, $n=16$, $k=4$) in an 8 GiB pool in (24) give 76 concurrent trees for APC and 103 for PD-Prune. Under a 1 s P99 TTFT constraint the same model admits 384 QPS for APC and 512 for the pruned tree, and saturated decode throughput rises from 48640 to 65920 tokens/s (+35.5%). This uses measured M , not the control-plane peak of Table 10. Figure 12 is that sweep.

C.5 GPU PD-Prune+

Setup. Qwen3-8B, bfloat16, $n=80$, $D=512$, $\text{tp}=2$ and 4 on four A100-SXM4 40GB GPUs. PD-

Table 14: Qwen3-8B GPU forest, $n=80$, $D=512$. Peak KV is independent of tp . Game24 / HumanEval peaks stay with the spine.

	peak	D^+	e2e	acc.	time reduction	G24	HE
APC $\text{tp}=2$	1582	40960	36.0	0.900	—	1289	1970
PD-Prune+ $\text{tp}=2$	1182	13356	28.2	0.863	0.78×	793	1882
APC $\text{tp}=4$	1582	40960	22.7	0.938	—	1289	1970
PD-Prune+ $\text{tp}=4$	1182	12826	16.9	0.925	0.75×	793	1882

Prune+ is the admission rule of §3.1 on the aliased trunk. GSM8K and Game24 are ToT ($k=4$); HumanEval is ReAct wrap+recovery ($k=2$).

Peak KV on this forest is 1182 against APC 1582 ($0.75\times$) at both widths. Game24 and HumanEval peaks are $0.62\times$ and $0.96\times$ APC.

C.6 Multi-turn Tree-of-Thoughts

Setup. Qwen3-4B and Qwen3-8B, bfloat16, $\text{tp}=2$, temperature 0, $n=16$. Each problem is one session of $k=4$ agents and three turns, and each system issues that problem as its own generate. Turns 1 and 2 decode a 32-token step; the losers abort; the winner spine is the parent of the next fan-out. Turn 3 is the published answer, $D=512$ on GSM8K and $D=256$ on Game24. On Game of 24 the three turns propose an equation, continue on the two cards that remain, and write the expression that uses all four cards. On GSM8K they set up the algebra, check or rewrite the previous step, and collect the final number. The prefill gate runs at every fan-out, before queue. The 32-token step is issued only to children that were admitted.

Occupancy across turns. Table 6 and Figure 6 are GSM8K and Game24 under that schedule. Peak KV is 275 against APC 318 on GSM8K ($0.86\times$) and 216 against 265 on Game24 ($0.82\times$),

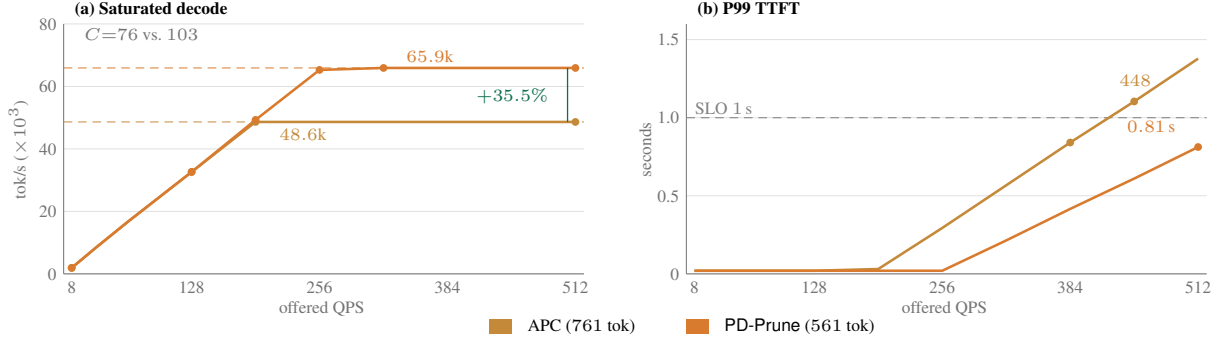


Figure 12: **SLO sweep** from the measured DeepSeek-R1 GSM8K peaks (761 APC tokens, 561 spine tokens, $n=16$, $k=4$). An 8 GiB pool at 147456 bytes/token holds 76 and 103 trees. Each slot serves 640 tokens/s, so offered load saturates at 48640 and 65920 tokens/s. Under the same model the 1 s P99 TTFT line is crossed by APC at 448 QPS and is still met by PD-Prune at 512 QPS (0.81 s).

Table 15: Expand versus decode, milliseconds per problem. Expand is the prefill of the k residuals across three turns. On APC that timer also includes the two 32-token step decodes; PD-Prune and PD-Prune+charge those steps to decode. Speedup in Figure 13 is APC expand time over the other system. Shortest time in each trio in green.

	Expand (ms)			Decode (ms)		
	APC	FS	FS+	APC	FS	FS+
4B GSM8K	429	83	55	1215	1680	1572
4B Game24	435	87	53	1166	1552	1074
8B GSM8K	597	94	58	1867	2468	1848
8B Game24	598	94	56	1538	2060	1466

on both models. PD-Prune+is the end-to-end reduction on this schedule: 1.91 s against APC 2.48 s on Qwen3-8B GSM8K, 1.52 s against 2.15 s on Game24, and 1.13 s against 1.61 s on Qwen3-4B Game24. Across 16 problems, three turns, and four agents, the expand issues 192 residuals. PD-Prune+prefills 48 of them and withholds the other 144 before they reach the GPU.

Table 15 and Figure 13 separate that expand from the winner decode. The expand speedup is $5.0\text{--}6.4\times$ for PD-Prune and $7.9\text{--}10.6\times$ for PD-Prune+. Decode does not match it. PD-Prune decode is slower than APC because the two step decodes are timed there. PD-Prune+is the column that shortens the trace: Qwen3-8B GSM8K falls from 2.48 s to 1.91 s, Qwen3-8B Game24 from 2.15 s to 1.52 s, and Qwen3-4B Game24 from 1.61 s to 1.13 s. GSM8K accuracy matches APC on Qwen3-4B (15/16 on the same problems) and is one problem lower on Qwen3-8B (15/16 vs. 16/16). On Game24, PD-Prune solves two more problems than APC on Qwen3-4B (8/16 vs. 6/16) and the same nine on Qwen3-8B; PD-Prune+solves 9/16 and 10/16.

C.7 Cross-model forest

Setup. Same four systems, $n=40$, $D=512$, $tp=2$ and 4, on Qwen3-4B, Qwen2.5-Math-1.5B, Qwen3-14B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B.

Peak occupancy. APC retains the shared trunk and the union of the residuals, which is the copy-on-write term $M_{\text{APC}} = M_{\text{CoW}} = b(L + \sum_i \ell_i)$ from (1). The tree retains only the committed spine $M_{\text{spine}} = b(L + \ell_*)$. The ratio of these two terms cancels b and equals $(L + \ell_*)/(L + \sum_i \ell_i)$, so it depends on k and the lengths (L, ℓ_*) rather than on the weights or the tensor width. When the residuals have similar length, the ratio simplifies to $(L + \ell)/(L + k\ell)$ and decreases as L/ℓ decreases. Figure 5 shows this pattern: a long trunk on GSM8K gives 0.72–0.75, and a shorter trunk on Game24 gives 0.57–0.61. Figure 14 repeats the comparison at $n=80$ with recompute and PD-Prune+on separate axes. Relative to recompute, which stores the cloned occupancy M_{clone} , the spine is $0.25\times$. PD-Prune+matches that ratio. Accuracy remains at the APC level, which indicates that the lower peak is a storage effect on the same committed answer. Figure 15 shows the per-item decode-length distribution on GSM8K and Game24.

HumanEval uses $k=2$, so the residual is a smaller fraction of the context and M_{spine} remains close to M_{APC} . Pass@1 is unchanged across recompute, APC, and PD-Prune.

C.8 Branching, pruning, and long traces

Setup. GSM8K, $n=16$, $D=512$, $tp=2$, on Llama-3-8B, Mistral-7B, and DeepSeek-R1-8B.

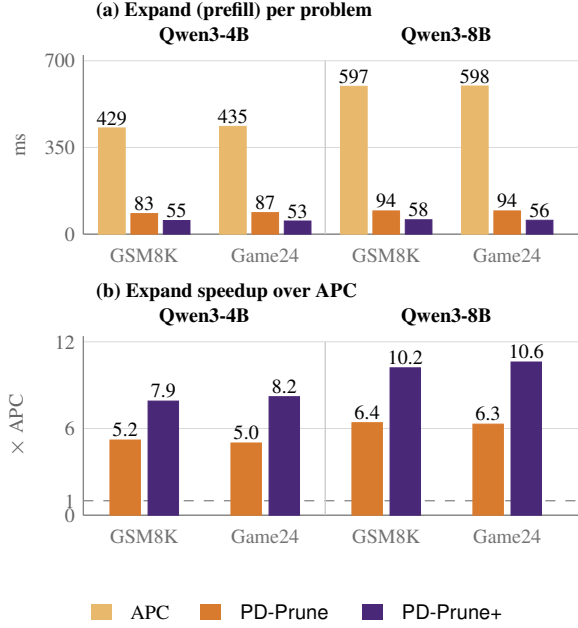


Figure 13: **Expand versus APC** on the same four runs as Table 15. (a) Expand milliseconds per problem. (b) Speedup of that expand over APC. The dashed line is $1\times$. PD-Prune is $5.0\text{--}6.4\times$ and PD-Prune+ is $7.9\text{--}10.6\times$. Decode in Table 15 does not repeat that ratio. PD-Prune+ is where end-to-end time falls, to $0.70\text{--}0.77\times$ APC on the three runs whose emitted tokens shrink.

The sweep changes k or adds a second ToT level. MATH-500 ($n=200$) and AIME ($n=73$) use $D=2048$.

The ratio falls with k . Table 17 and Figure 16 measures $(L + \ell)/(L + k\ell)$ on the GPU. The spine peak is invariant in k while the APC peak increases, so the ratio moves from $0.89\text{--}0.90$ at $k=2$ to $0.53\text{--}0.55$ at $k=8$ on all three families. At depth 2 the ratio is $0.75\text{--}0.76$. PD-Prune+ leaves the peak unchanged. At $k=8$ it rejects 28 branches and reduces fan-out from 307 to 179 ms on Llama-3-8B, from 349 to 191 ms on Mistral-7B, and from 363 to 194 ms on DeepSeek-R1. With draft pruning disabled, Llama-3-8B fan-out is 247 ms and DeepSeek-R1 fan-out is 271 ms. A batch of 16 keeps the ratio at $0.72\text{--}0.73$ ($2071/2871$ on Llama-3-8B, $2135/2935$ on Mistral-7B, $2093/2893$ on DeepSeek-R1).

At $D=2048$ the ratio is $0.75\text{--}0.91$ (Table 18). PD-Prune+ cuts DeepSeek-R1 MATH-500 from 429 s to 247 s and Qwen3-4B from 329 s to 236 s; the Qwen3-4B $n=32$ slice is 38 s versus 53 s. PD-Prune+ cuts MATH-500 fan-out from 2406 to 1284 ms on DeepSeek-R1 and from 2787 to

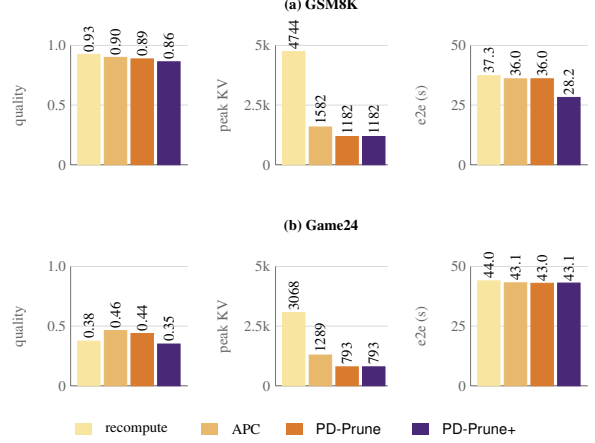


Figure 14: **Four systems on Qwen3-8B** ($n=80$, $tp=2$, $D=512$). Each panel has its own axis. Quality is higher-better; peak KV and end-to-end time are the measured values, so a shorter bar is better. (a) GSM8K. (b) Game24. Figure 14 shows that PD-Prune and PD-Prune+ lower peak KV below APC on both workloads ($0.75\times$ on GSM8K, $0.62\times$ on Game24) while GSM8K quality stays with APC, and that PD-Prune+ also lowers GSM8K end-to-end time from 36.0 s to 28.2 s.

1014 ms on Qwen3-4B, against APC at 1593 and 1201 ms. Qwen3-4B PD-Prune+ scores 120/200 on MATH-500 (APC 124/200).

D Extended Related Work

Inference-time search. Chain-of-thought prompting showed that intermediate steps improve multi-step reasoning [56]. Tree-of-Thoughts and related graph search expand several of those steps, score them, and backtrack [45, 51]. Self-consistency samples many complete traces and keeps a majority answer [40]. The accuracy gain scales with the number of traces, and so does the token cost. EquivPruner identifies the redundancy that remains after naive widening: textually distinct steps can be the same mathematical action, and exploring every member of an equivalence class spends tokens without adding a new state [60]. PD-Prune does not replace the search procedure and does not decide semantic equivalence. It takes the residuals the harness has already proposed and decides, from the residual text and the radix index, which of them are prefilled and which pages cross the prefill-decode connector.

Path pruning. Decode-time pruners drop a candidate after generation has started. Beam search in Tree-of-Thoughts scores a finished step and keeps the top beams [45]. ESC draws a window of com-

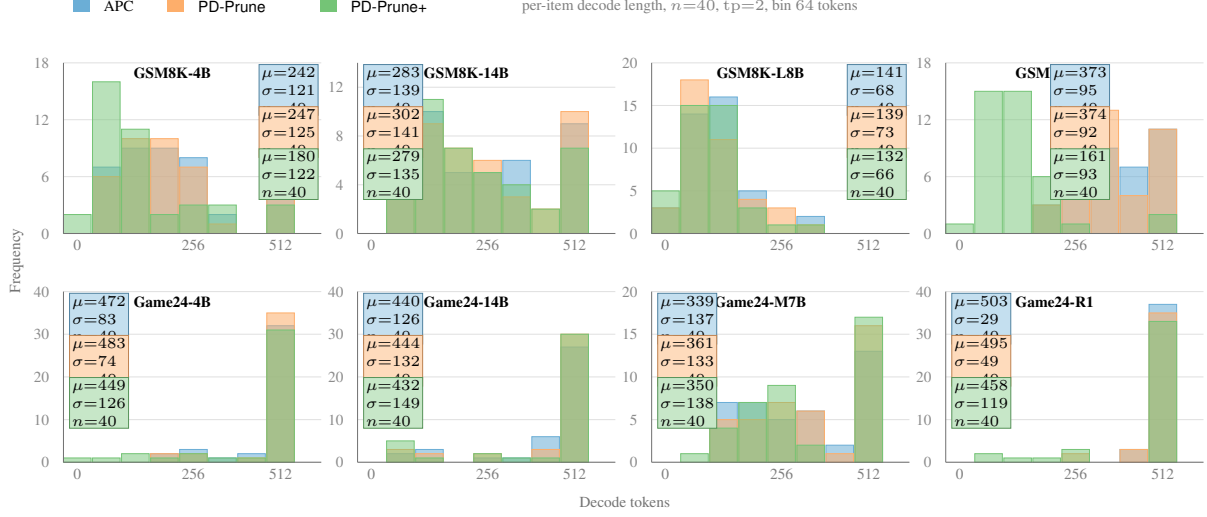


Figure 15: **Per-item decode-length histograms** ($n=40$, $tp=2$, $D=512$). Each panel overlays APC, PD-Prune, and PD-Prune+ (bin width 64). Boxes report μ , σ , and n . GSM8K (top) and Game24 (bottom) overlay APC, PD-Prune, and PD-Prune+. Llama-3-8B Game24 is omitted (all lengths 1); Mistral-7B is shown instead.

Table 16: Spine peak versus an optional last-line decode stop, $n=32$, $tp=2$. PD-Prune+ is the prefill admission stack on the aliased trunk: hash skip, draft prune, residual-prefix abort, and the connector gate. It keeps the same spine as the alias row and does not stop decode. This slice also ends committed decode when the last non-empty line is a finished answer; that processor is separate, and it is what moves accuracy and D^+ . Peak is the max over chunks. GSM8K $D=512$; Game24 $D=256$. R1-8B is DeepSeek-R1-Distill-Llama-8B; L8B is Llama-3-8B; M7B is Mistral-7B. Best storage in green.

Model	Metric	APC	FS+
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
L8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
M7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

plete samples and skips later samples only once that window agrees [21]. Speculative Rejection ranks partial generations and halts the lower quantile after a partial-reward horizon [31]. DPTS generates a fixed mini-step on every active node, then early-stops low-confidence children [19]. STOP gives these methods a common taxonomy, by whether the signal is internal or external and whether it is learned, and shows that a learned internal token can prune a path from a short prefix [61]. EquivPruner prunes at action granularity: a lightweight equiva-

Table 17: GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). The spine peak is invariant in k ; the APC peak increases with k . ratio is FS/APC. L8B is Llama-3-8B; M7B is Mistral-7B; R1-8B is DeepSeek-R1-Distill-Llama-8B. Best storage in green.

Model/ratio	Method	$k=2$	$k=4$	$k=8$	depth 2
L8B	APC	610	750	998	814
	FS	550	550	550	614
FS/APC		0.90	0.73	0.55	0.75
M7B	APC	629	769	1033	841
	FS	569	569	569	641
FS/APC		0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	FS	561	561	561	641
FS/APC		0.89	0.74	0.53	0.76

lence detector, trained on mathematical statement pairs, retains one representative of each equivalent set during search [60]. The shared limitation is the input to the score. ESC, Speculative Rejection, DPTS, and STOP read tokens that already reside in the KV cache; EquivPruner compares candidate steps that have already been written. PD-Prune withholds a repeated loop before the prefill kernel and continues every other low score past a t_0 -token probe only when the generated prefix meets the decode threshold (§3.2). The gate is host-agnostic. §C.3 places the same τ_{pre} in front of ESC, Speculative Rejection, and DPTS, and the prefill column moves while the host’s later cut still applies. A learned equivalence or early-stop signal can sit on top of that gate; the experiments treat the host as the decode policy and the residual score as the prefill policy.

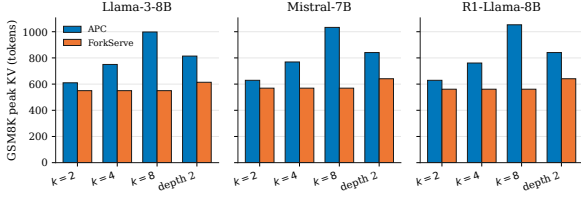


Figure 16: **GSM8K peak KV versus branching** ($n=16$, $tp=2$, $D=512$). PD-Prune+matches the PD-Prune bar. The spine peak is invariant in k ; the APC peak grows with the hashed union, so the ratio falls from 0.89–0.90 to 0.53–0.55.

Table 18: Contest forest, $D=2048$, $k=4$, $tp=2$. MATH-500 is $n=200$; AIME is $n=73$. R1 is DeepSeek-R1; 4B is Qwen3-4B. Best storage and shortest end-to-end time in green.

Model	Method	peak	/ APC	acc.	e2e (s)
R1 MATH	APC	1863	—	0.560	429
	FS	1391	0.75	0.510	432
	FS+	1599	0.86	0.545	247
R1 AIME	APC	2118	—	0.055	178
	FS	1646	0.78	0.123	178
	FS+	1662	0.78	0.096	165
4B MATH	APC	1937	—	0.620	329
	FS	1770	0.91	0.625	248
	FS+	1770	0.91	0.600	236
4B AIME	APC	2150	—	0.027	136
	FS	1794	0.83	0.041	136
	FS+	1794	0.83	0.041	131

Request-level serving. Orca introduced iteration-level scheduling [47]. vLLM paged the KV cache and hashes full blocks for automatic prefix caching [17]. SGLang added radix prefix sharing for structured programs [53]. Sarathi chunked pre-fills so a decode batch can share the GPU with a prompt [2, 3]. DistServe and Splitwise disaggregated prefill from decode and moved KV across a connector [54, 32]. Prefix caching discovers sharing after tokens exist, which is why a same-batch fan-out still clones the trunk. Disaggregation transfers every page the prefiler produced, which is why an aborted residual still occupies the connector. (4) uses the hash as the page-aligned match m_i and the connector as X_i : both are instances of one residual action, and the trunk alias removes L from both.

Program- and SLO-aware scheduling. Autellix schedules by program-level attained service [25]. JITServe allocates just-enough bandwidth under heterogeneous SLOs [52]. Hermes, ThunderAgent, and SAGA raise the scheduling unit to the program or workflow [24, 16, 13]. These systems leave KV unshared across children of one fan-out and leave unissued children unpre-filled until the program reaches them. PD-Prune is complementary: the program still chooses the search, and the engine

admits each named residual once, against the pages already published.

KV retention, overlap, and cache edits. InferCept, Continuum, MORI, and ConServe keep the current context alive or placed across tool gaps [1, 20, 43, 11]. Sutradhara overlaps tool execution with prefill of one linear tool-independent prefix [6]. SmoothAgent lookahead-transforms context; IdleSpec drafts plans for accuracy rather than for KV reuse [30, 9]. ForkKV copy-on-write splits LoRA residuals across agents that share a base cache [38]. CachePrune edits KV neurons so the model does not follow instructions embedded in retrieved context [62]. That edit changes which attention state is visible inside one request. It does not decide which sibling residual is prefilled, and it does not change the connector payload of a fan-out. (4) aliases the shared prefix, then sets W_i and X_i from the unmatched suffix.

Speculative decoding. Classical speculative decoding and multi-head drafters propose *output* tokens and verify them against the target model [18, 7, 22]. The object stored by PD-Prune is a future prompt prefix: the residual the harness will observe, not a draft of the tokens the model will emit. Admission stores that prefix under an admit probability and a prefix-hit rate (§B.3). Speculative tokens stay outside the sampler, and (8) binds the prompt that actually arrives.